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Test – 23

Science & Tech / Environment Revision

Topics covered:

- Bio Technology + IPR
- Health
- Energy, Research and Development
- Environment Revision
- Current Affairs September 2021



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DEDICATED PRELIM PROGRAM(DPP)

UPSC-CSE PRELIMINARY TEST SERIES – 2021/2022

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TEST BOOKLET SERIES

Test Code:

DPP

**TEST BOOKLET
GENERAL STUDIES
TEST - 23**

E 23

Time Allowed: Two Hours

Maximum Marks : 200

INSTRUCTIONS

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET DOES NOT HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC.

2. ENCODE CLEARLY THE TEST BOOKLET SERIES CODE A, B, C OR D AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.

3. You have to enter your Roll Number on the Test Booklet in the provided alongside this line. DO NOT write anything else on the Test Booklet.

4. This Test Booklet contains 100 questions. Each question is printed in English. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose ONLY ONE response for each item.

5. All items carry equal marks.

6. Penalty for wrong answers:

THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BY A CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.

- I. There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, One-Third of the marks assigned to that question will be deducted as penalty.
- II. If a candidate gives more than one answer, it will be treated as a wrong answer even if one of the given answers happens to be correct and there will be same penalty as above to that question.
- III. If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

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TOPICS COVERD: Bio Technology + IPR, Health, Energy, Research and Development, Environment Revision, Current Affairs September 2021



Q.1) Match list 1 with list 2 and select the correct answer using the code given below the lists:

List-1 (river)

list-2(dams)

- | | | |
|-------------|---|---------------------|
| A) Godavari | - | 1. gandhisagar |
| B) Chambal | - | 2. sriram sagar dam |
| C) Banas | - | 3. idukki dam |
| D) Periyar | - | 4.dantiwada dam |
- a) A-2, B-1, C-4, D-3
b) A-3, B-1, C-3, D-4
c) A-4, B-1, C-4, D-4
d) A-3, B-2, C-4, D-3

Q.2) Consider the following statement:

1. Bhagat Singh was a revolutionary leader who believed in socialist, secular and democratic India.
2. He was a member of 'Hindustan republican association (HRA)'.
3. He started a youth organization called 'young India league'.
4. He wrote an essay 'Why I am atheist'.

Which of the statements given above is/are correct?

- a) 1 and 2
b) 1, 2 and 4
c) 3 and 4
d) 1 and 4

Q.3) consider the following statement about Euro green bond:

1. It is the first ever euro denominated green bond issuance from india.

2. A green bond is a type of debt instrument that proceeds of a green bond offering are earmarked for use towards financing green projects.

Which of the Statements given above is/are correct?

- a) 1 only
b) 2only
c) Both 1 and 2
d) Neither 1 nor 2

Q.4) Consider the following statement AUKUS alliance:

1. It is a security alliance of Australia, USA and UK will focus on advancing strategic interest in the Indo-pacific region.
2. Australia will become just the fifth nation in the world to operate nuclear powered submarines after the US, UK, China and Russia.

Which of the statements given above is/are correct?

- a) 1 only
b) 2 only
c) Both 1and 2
d) Neither 1 nor 2

Q.5) Consider the following statements:

1. SDR is a currency which is defined and determined by the international Monetary Fund (IMF).
2. The currency value of SDR is determined by summing the values in US dollars.
3. The SDR basket of currencies includes the US dollar, Euro, Japanese yen, Indian rupees and Chinese renminbi.

Which of the statements given above is/are correct?

- a) 1 and 2 only
b) 2 only
c) 2 and 3 only



d) 3 only

Q.6) With reference to National Mission on Cultural Mapping (NMCM), consider the following statement:

1. NMCM has now been handed over to the Indira Gandhi National center for art.
2. The entire project would take five years and a budget of ₹89 crore had been approved for it.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.7) 'SPIN' scheme, which was launched recently, aims at-

- a) sustainable environment by recycling plastic waste in the marine ecosystem.
- b) preventing child labour
- c) sustainable development by creating local self-employment.
- d) to regulate betting games.

Q.8) Recently, the India led resolution called 'Resolution 2593' adopted by the United Nations security council is related to which of the following organisations?

- a) Al-Qaida
- b) Taliban
- c) Lashkar-e-taiba
- d) Houthi rebels

Q.9) Consider the following statements with reference to the Account Aggregator:

1. It is a regulated entity that helps an individual securely and digitally access and share information from one to another financial institution in the Account Aggregator network.
2. Registering with an Account Aggregator is mandatory for customers.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.10) Consider the following statements with reference to the World Heritage Institute of Training and Research –Asia Pacific:

1. It is a non-profit organization specialized in the area of heritage conservation under the auspices of the World Bank.
2. Mission of the institution is to strengthen implementation of the World Heritage Convention 1972 in the Asia and the Pacific region

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.11) Consider the following statements with reference to the Ubreath Life :

1. The Ubreath Life is the world's first, state-of-the-art 'Smart Bio-Filter' that can purify the air.
2. The technology used in this product is the 'Urban Munnar Effect' to amplify the phytoremediation process of the plants

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.12) Which of the following statements is incorrect with respect to the International Fund for Agricultural Development?

- a) It is a specialized international financial institution of the United Nations working in the field of poverty eradication.
- b) It was created in 1977, which is the outcome of the World Food Conference of 1974.
- c) It invests in rural people, empowering them to increase their food security, improve the nutrition of their families and increase their incomes.
- d) It's Headquarter is situated in Paris.

Q.13) Consider the following statements with reference to the Appointment of judges :

1. Articles 123 of the Indian Constitution govern the appointment of judges to the Supreme Court (SC)



2. Judges of the higher judiciary are appointed only through the collegium system.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.14) Consider the following statements with reference to the Nipah Virus:

- 1. Nipah Virus is a zoonotic virus as it is transmitted from animals to humans.
- 2. There are no vaccines for treatment of both humans and animals.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.15) In the context of economy , the term "Animal Spirits" refers to

- a) the free market helps the economy grow.
- b) human emotions and instincts can drive financial decision-making.
- c) regulations by state can be useful to the economy.
- d) None of the above.

Q.16) Which one of the following is remembered for the Swadeshi Steam Navigation Company to challenge the monopoly of the British in sea trade?

- a) Bal Gangadhar Tilak
- b) V. O. Chidambaram Pillai
- c) Mahatma Gandhi
- d) Dheeran Chinnamalai

Q.17) Consider the following statements with reference to the Bitcoin:

- 1. It is the first decentralized peer-to-peer payment network that is powered by its users with no central authority or middlemen.
- 2. Singapore became the first country in the world to adopt bitcoin as legal tender.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.18) Consider the following statements with reference to the Common Services Centre (CSC) programme:

- 1. It is an initiative of the Ministry of Electronics & IT.
- 2. E-Governance Services India Limited is a Special Purpose Vehicle to monitor the implementation of the Common Services Centers Scheme.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.19) Which of the following statements is incorrect with respect to the Fragile X Syndrome (FXS)?

- a) It is a genetic disorder that causes a range of developmental problems including learning disabilities and cognitive impairment.
- b) It is the leading inherited cause of autism in 4% of the global population.
- c) It is caused by the mutation named FRAXA.
- d) The simplest tool for timely detection is a RNA test.

Q.20) Which of the following statement is incorrect regarding Raja Mahendra Pratap Singh:

- a) Raja Mahendra Pratap Singh established provisional govt of India at Bagh-e -Babur in Kabul in the middle of world war I 1915
- b) He launched the world federation in America moreover he was nominated for the 1945 nobel peace prize.
- c) He returned to India a year before Independence and immediately began work with Mahatma Gandhi.
- d) He won Lok Sabha elections as an Independent candidate from Mathura in 1957.

Q.21) "Extends over central and southern Europe, eastern North America and western China. Moderate average temperature and abundant rainfall in this region, the flora includes trees like beech, oak, maple and cherry. Most animals are the familiar vertebrates and invertebrates, these are generally the most productive agricultural areas of the Earth." This is the most likely description of -

- a) Tropical rainforest
- b) Temperate deciduous forest
- c) Dry deciduous forest
- d) Coniferous forest



Q.22) Consider the following statements regarding 'Vidyanjali 2.0':

1. It is an initiative taken by the Ministry of Social Justice and Empowerment.
2. It has three verticals: "Participate in school Service/Activity", "Helping hand of India" and "Assets/Material/Equipment" in which volunteers can support and strengthen the government and government aided schools.

Which of the statements given above is/are correct?

- a) 1 Only
- b) 2 Only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.23) Recently launch of the CRISP-M tool helps in

- a) Climate risk management.
- b) Gene editing.
- c) Water management.
- d) Weather predictions.

Q.24) With reference to the Karuppur kalamkari painting, consider the following statements:

1. It is a hand-painted cotton textile in North East India.
2. Recently it has received the geographical indication (GI) tags.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.25) Consider the following statements with regard to the Kunming Declaration:

1. It calls upon the parties to "mainstream" biodiversity protection in decision-making and recognise the importance of conservation in protecting human health.
2. The theme of the declaration is Ecological Civilization: Building a Shared Future for All Life on Earth.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.26) Consider the following statements regarding financial stability and development (FSDC) council:

1. FSDC is a statutory apex council under the Ministry of economic affairs.
2. Account aggregator is the initiative of FSDC.
3. Union Minister of Finance is the ex-officio chairman of FSDC.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 3 only

Q.27) Which of the following indices / reports are published by the UN Environment programme (UNEP)?

1. Global environment outlook.
2. Climate change performance index.
3. State of global climate report.
4. Emissions gap report.
5. Living planet report

Select the correct answer using the code given below.

- a) 1 and 4 only
- b) 1 and 5 only
- c) 2, 3 and 5 only
- d) 1,4 and 5 only

Q.28) Consider the following statement:

1. Convention on biological diversity, a legally binding agreement, covers all the ecosystem, species and genetic resources.
2. Convention on the conservation of migratory species (CMS) is the only global conversation on specializing in the conservation of migratory species.
3. Intergovernmental Panel on Climate Change (IPCC) is jointly established by the world meteorological organization and Vienna convention.
4. TRAFFIC is a joint conservation programme worldwide for nature (WWF) and international union conservation and nature.

Which of the statements given below is/ are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1, 2 and 4 only
- d) 1,2,3 and 4

Q.29) Which of the following statements is/ are correct regarding the animal welfare board of India?



1. The Animal welfare board of India is a Non-profit organization on animal welfare laws and promotes animal welfare in the country.
2. Give financial assistance and other assistance to animal welfare organisations, functioning in any local areas or to encourage the formation of animal welfare organisations.

Select the correct answer using the code given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2 only
- d) Neither 1 nor 2

Q.30) Consider the following statement:

1. Nilgiri tahr is naturally found in India only.
2. Red pandas are naturally found in India only.
3. Kashmir stag is naturally found in India only.
4. Indian buffalo are naturally found in India only.

Which of the statements given above is/ are incorrect?

- a) 1 and 2 only
- b) 2 and 4 only
- c) 1,2 and 3 only
- d) 1,2,3 and 4

Q.31) Consider the following statements with reference to the Snow Leopard:

1. The Snow Leopard acts as an indicator of the health of the mountain ecosystem in which they live.
2. It is protected under Schedule I of Indian Wildlife (Protection) Act 1972.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.32) Which of the following is best described as Phytoremediation?

- a) It is a process by which plants effectively remove pollutants from the air.
- b) It is a process by which microorganisms effectively remove pollutants from the air.
- c) it is the use of microbes to clean up contaminated soil and groundwater.
- d) None of the above.

Q.33) Consider the following pairs with reference to conventions and their purposes:

Conventions

1. Carpathian Convention
2. The Bamako Convention
3. The Tehran Convention

Purposes

- Protection and Sustainable Development of the Carpathian Mountain Range
- African nations prohibit the import of any hazardous waste.
- Environmental protection and sustainable development in the Caspian Sea region.

Which of the above pairs is/are correctly matched?

- a) 1 and 2 only
- b) 2 only
- c) 3 only
- d) 1, 2 and 3

Q.34) Climate Change Performance Index (CCPI) is released by -

- a) United Nations Environment Programme (UNEP)
- b) Germanwatch, New Climate Institute and the Climate Action Network.
- c) World Wildlife Fund (WWF)
- d) The Intergovernmental Panel on Climate Change (IPCC)

Q.35) 'Assessment of Climate Change over the Indian Region' is released by -

- a) Indian Space Research Organization
- b) Ministry of Earth Sciences
- c) Aryabhata Research Institute of Observational Science
- d) The Ministry of Environment and Forests

Q.36) With reference to 'SEaweEDS', consider the following Statements:

1. They are macroscopic algae and have no differentiation of tissues such as roots, stems and leaves.
2. Seaweeds grow in shallow coastal waters.
3. Seaweeds are used as a drug for goiter treatment, intestinal and stomach disorders.

Which of the Statements given above are correct ?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only



d) 1, 2 and 3

d) 2, 3 and 4 only

Q.37) Consider the following Statements:

1. Conservation of Biodiversity
2. Recharge of aquifers
3. Decrease in bird and animal population
4. Soil conservation

Which of the above statements is / are correct about the ecological significance of 'Sacred Groves' in India ?

- a) 1 and 2 ,4 only
- b) 2 and 3 ,4 only
- c) 1 and 3 only
- d) All the above

Q.38) Consider the following statements with regard to the animal Dhole:

1. They are native to Central , South , East Asia, and Southeast.
2. It is protected as Schedule I species under the Wildlife Protection Act, 1972.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.39) Consider the following statements with regard to the Central Zoo Authority:

1. It was constituted in 1992 under the Wildlife (Protection) Act, 1972.
2. It is chaired by the Environment secretary and has 10 members.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.40) Consider the following :

1. Aerosol
2. Volatile organic compounds
3. Black carbon
4. Oxides of nitrogen.

Which of the above are released into the atmosphere due to the burning of crop/ biomass residue?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1, 2 and 4 only

Q.41) As per the Bio- Medical Waste management Rules in India, which one of the following statements is correct.

- a) Biochemical waste comprises only human anatomical waste, treatment apparatus.
- b) Biochemical waste segregation into 10 categories and packs 5 colour backs for disposal
- c) The quantum of waste generated in India is estimated to be 20-30 kg per day in a hospital and 40 kg per day in the bed of a clinic.
- d) None of the above

Q.42) Consider the following statement :

1. Faster Adoption and manufacturing of (Hybrid) and Electric Vehicle schemes will help promote use of electric and hybrid vehicles.
2. Carbon sequestration the technologies designed tackle global warming by capturing NO₂ and SO₂ at power stations, industrial sites or even directly from the air and permanently storing it underground.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.43) Which of the following statements is correct about the Plastic Waste Management Amendment Rules 2021?

1. The manufacture, import, stocking, distribution, sale and use of the identified single-use plastic will be prohibited with effect from the 1st July, 2022.
2. The permitted thickness of the plastic bags, currently 100 microns, will be increased to 175 microns from 30th September, 2021, and to 20 microns from the 31st December, 2030.

Select the correct answer using codes given below.

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.44) National Park lies partly in Golaghat District and partly in Nagaon District of Assam. It is the oldest park in Assam covering an area of 430 sq km along the river Brahmaputra in the North and the Karbi Anglong hills



in the South. It is a world heritage site that is famous for the Great Indian one-horned rhinoceros. It is recognized as an Important Bird Area by BirdLife International for the conservation of avifaunal species.

Which one of the following National Parks is described in the above passage?

- a) Raimona national park
- b) Kaziranga national park
- c) Manas National park
- d) Orang national park

Q.45) Consider the following statements regarding the difference between a national park and a biosphere reserve in India:

- 1. While a national park is created to protect the habitat of a species, a biosphere reserve aims to protect an entire ecosystem.
- 2. While human activities such as forestry and grazing are permitted in a national park, they are prohibited in a biosphere reserve.

Which of the statements given above is/are correct?

- a) 1 Only
- b) 2 Only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.46) Consider the following statement :

- 1. An oil spill refers to any uncontrolled release of crude oil, gasoline, fuels, or other oil by-products into the environment.
- 2. Compostable plastics are derived from non-renewable materials like corn, potato, and tapioca starches, cellulose, soy protein, and lactic acid.
- 3. Bioindicators are non-living organisms such as plants, plankton, animals, and microbes, which are used to assess the health of the natural ecosystem in the environment.
- 4. Bioremediation is the use of microorganisms to degrade the environment contaminants into less toxic forms.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 1 and 4 only
- c) 2, 3 and 4 only
- d) 1, 2, 3 and 4

Q.47) Consider the following statement with reference to Hydrofluorocarbons (HFCs).

- 1. HFCs are entirely natural.

- 2. It's used in refrigeration, air-conditioning, insulating foams and aerosol propellant.
- 3. It was developed to replace stratospheric ozone-depleting substances (ODS).

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2 and 3

Q.48) Consider the following statement:

- 1. The Hornbill Festival celebrated in Gujarat is named after the bird – Hornbill, which is the most revered and admired bird for the Gond tribe.
- 2. Dragonflies are insects that act as important bio-indicators of the ecological health of an area.
- 3. The tiger survey is usually held once every four years and elephants are counted once every five years.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 1, 2 and 3 only

Q.49) Which of the following statements regarding a grazing food chain is/are correct?

- 1. In all grazing food chains the first trophic level is always occupied by producers.
- 2. Herbivores are always placed at the third trophic level in a grazing food chain.

Select the correct answer using the code given below.

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.50) The world energy outlook report is released by which of the following?

- a) International atomic energy agency
- b) International renewable energy organization
- c) International Nuclear energy organization
- d) International energy agency

Q.51) With the reference to black carbon consider the following statement:

- 1. Black carbon is a short-lived pollutant.
- 2. It is the fifth largest contributor to warming the planet after methane.



3. It absorbs solar energy and warms the atmosphere.
4. It darkens the surface of snow ice thus, increasing the albedo of snow.

Which of the statements is/ are not incorrect?

- a) 2 and 4 only
- b) 1 and 3 only
- c) 1,2 and 3 only
- d) 1,2,3 and 4

Q.52) Consider the following statement:

1. Saltwater crocodile is found in Odisha's Bhitarkanika National Park, the Sundarbans in West Bengal.
2. Mahseer is found only in India.
3. Gharial is a species of Asian crocodilian distinguished by their long, thin snouts.
4. A tree with an estimated age of 10 years or more shall be defined as a heritage tree.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 2 and 4 only
- c) 1,2 and 4 only
- d) 2, 3 and 4 only

Q.53) Consider the following statements about Global Assessment Report (GAR) on Drought 2021:

1. Global assessment report (GAR) on Drought 2021 Released by UN Office for Disaster Risk Reduction (UNDRR).
2. The report explores the systemic nature of drought and its impacts on the achievement of the Sendai Framework for Disaster Risk Reduction, the SDGs, and human and ecosystem health and wellbeing.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.54) "It is one of the very few mammals that build its own home, or nest, complete with a 'roof'. It is also an indicator species. Its presence reflects the health of its primary habitat, the tall, wet grasslands of the region. Once found along a narrow strip of tall and wet grassland plains on the Himalayan foothills."

This is most likely the description of -

1. Indian peacock
2. pygmy hog
3. Himalayan tahr

4. Mishmi takin

Q.55) With reference to the United Nations Convention to Combat Desertification(UNCCD) consider the following statement.

1. It is the sole legally binding international agreement linking environment and development to sustainable land management.
2. The Convention addresses specifically the wetlands and grassland areas, known as the dryland, where some of the most vulnerable ecosystems and peoples can be found.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.56) Which of the following are the powers of the central government under the environmental (protection) act, 1986?

1. It can set the standards for the emissions of pollutants from any source.
2. It can lay down the procedures and safeguards for the handling of hazardous substances.
3. It can compel a state government to take prior approval for directing any forest land for non-forest purposes.

Select the correct answer using the code given below.

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1,2 and 3

Q.57) "Located in the Himalayan foothills. UNESCO Natural World Heritage Site, a Project Tiger reserve, an elephant reserve and a biosphere reserve. The park is known for its rare and endangered endemic wildlife such as the hispid hare, golden langur and pygmy hog and famous wild water buffalo", is most likely the description of:

- a) Nameri Devi national park
- b) Manas National park
- c) orang national park
- d) Raimona national park

Q.58) Consider the following pairs with wildlife sanctuaries with their location.

Wildlife sanctuary	Location
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1. Bornadi Wildlife - Assam Sanctuary
2. Ramgarh Vishdhari wildlife sanctuary - Rajasthan
3. Periyar wildlife sanctuary - Karnataka
4. Dandeli wildlife sanctuary - Kerala

Which of the pairs given above is/are correct?

- a) 1 and 4 only
- b) 3 and 4 only
- c) 1 and 2 only
- d) 1,2 and 3 only

Q.59) Which of the following are in Nilgiri biosphere reserves ?

- a) Neyyar, peppara wildlife sanctuary, Bandipur and mukurthi national park, Sathyamanglam wildlife sanctuary.
- b) Mukurthi , Bandipur, Nagarhole, silent valley national park and Sathyamangalam, Wayanad wildlife sanctuary.
- c) Bandipur, Nagarhole, silent valley national park and papikonda wildlife sanctuary, kalakad mundanthurai wildlife sanctuary.
- d) shendurney wildlife sanctuary Sri Venkateswara wildlife sanctuary, Bandipur, Nagarhole, silent valley national park and Sathyamangalam, Wayanad wildlife sanctuary.

Q.60) Consider the following statements with reference to the Mission Innovation:

1. Mission Innovation (MI) is a global initiative of 30 countries and the European Commission to make clean energy affordable, attractive and accessible for all.
2. This will accelerate progress towards the Paris Agreement goals and pathways to net zero.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.61) Which of the following statements is incorrect with respect to the Clean Energy International Incubation Centre?

- a) The Clean Energy International Incubation Centre (CEIIC), is a joint initiative of Tata Trusts and the Government of India.
- b) It has been set up for promoting innovations in the energy space

- c) It became the second International Incubator in India under Mission Innovation.
- d) It focuses on three core areas for innovation Energy Access, Energy Efficiency and Breakthrough Clean Technology.

Q.62) Consider the following statements with reference to the Clean Energy Research Initiative :

1. Clean Energy Research Initiative is a technology Development Scheme.
2. It helps to gain value for such collaborations and also develop a critical mass of researchers to meet requirements of R&D professionals for clean energy.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.63) Consider the following statements with reference to the Partnership to Advance Clean Energy :

1. It is launched by the United States and India to accelerate inclusive, low carbon growth by supporting research and deployment of clean energy technologies
2. The activities of the PACE initiative are aligned to support India's ambitious clean energy targets.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.64) Consider the following statements with reference to the Lignocellulose Hydrolysis Technology:

1. Lignocellulosic biomass can be utilized to produce ethanol, a promising alternative energy source for the limited crude oil.
2. The cost of ethanol production from lignocellulosic materials is relatively low based on current technologies

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2



Q.65) Consider the following statement regarding New fly ash utilization rules:

1. Fly ash notification was issued in 1999 to ensure 70% fly ash utilization in India by 2005.
2. The non-complying power plants will provide ash free-of-cost to agencies engaged in construction activities within a 300 km radius with all transportation costs to be borne by Thermal power plants.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.66) With reference to the DNA Technology (Use and Application) Regulation Bill, 2019, which of the following statements is/are correct?

1. The DNA Technology (Use and Application) Regulation Bill, 2019" is for expanding the application of DNA-based forensic technologies to support and strengthen the justice delivery system of the country.
2. The Bill mandates the permanent storage of DNA data obtained under all categories, including those obtained from a criminal investigation.
3. It also has the provisions to establish National and Regional DNA Data Banks for the maintenance of the national forensic DNA database for the purposes of identification of the aforementioned category of persons.

Select the correct answer using the code given below:

- a) 1 only
- b) 2 and 3 only
- c) 3 only
- d) 1, 2 and 3

Q.67) The gene BBX11 used to facilitate -

- a) bioremediation.
- b) greening of plants.
- c) cure of diabetes.
- d) fast growth of animals.

Q.68) Which of the following statements is incorrect with respect to the Genetically Modified Crops ?

- a) Genetic modification is a technology that involves inserting DNA into the genome of an organism.
- b) The seeds produced by these plants will inherit the new DNA.

- c) The Genetically Modified Crops permitted for cultivation in India are Bt Cotton and Bt Brinjal.
- d) Genetic Engineering Appraisal Committee (GEAC) is the apex body that allows for commercial release of Genetically Modified Crops.

Q.69) Consider the following statements with reference to the Genetic Engineering Appraisal Committee (GEAC) :

1. The Genetic Engineering Appraisal Committee (GEAC) functions in the Ministry of Science and Technology.
2. It is responsible for appraisal of activities involving uses of hazardous microorganisms and recombinants in research and industrial production from the environmental perspectives.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.70) Consider the following statements with reference to the Nanobiotechnology:

1. Nanobiotechnology is a discipline in which tools from nanotechnology are developed and applied to study biological phenomena.
2. Nanobiotechnology enables technology relevant for diverse sectors, such as chemicals, consumer products, health, energy, various other industries and the environment.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.71) Which of the following statements is incorrect with respect to the Genome editing Technology ?

- a) These technologies allow genetic material to be added, removed, or altered at particular locations in the genome.
- b) Several approaches to genome editing have been developed one-of them is known as CRISPR-Cas9, which is short for clustered regularly interspaced short palindromic repeats.
- c) CRISPR-Cas9 was adapted from a naturally occurring genome editing system in viruses.
- d) None of the above



Q.72) Consider the following statements with reference to the Stem cells :

1. Stem cells are the body's raw materials cells from which all other cells with specialized functions are generated.
2. These cells come from Embryonic stem cells and not Adult stem cells.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.73) Which of the following statements is incorrect with respect to the Biotech-KISAN ?

- a) It is an initiative of the Ministry of Agricultural and Farmers welfare.
- b) It aims to understand the problems of water, soil, seed and market faced by the farmers and provide simple solutions to them.
- c) Linking available science and technology to the farm for providing solutions to those problems.
- d) It aims to work with small and marginal farmers for better agriculture productivity

Q.74) Consider the following statements with reference to the Human Microbiome:

1. The Human Microbiome is the collection of all the microorganisms living in association with the human body.
2. These microbes are generally not harmful to us, in fact they are essential for maintaining health.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.75) Which of the following statements is incorrect with respect to the Patents (Amendment) Rules, 2021 ?

- a) The fastest granted patent is the one which was granted in 41 days after filing of such request.
- b) With regard to Initiatives for Startup India program, 80% fee concession has been provided for Patent applications filed by Startups.
- c) The Patent Prosecution Highway (PPH) is a set of initiatives for providing accelerated patent prosecution procedures by sharing information between some patent offices.

- d) This facility of Expedited Examination system is provided only for patent applications filed by Startups.

Q.76) Recently, International Intellectual Property Index is released by -

- a) World Trade Organisation
- b) US Chamber of Commerce
- c) World Intellectual property Organisation
- d) None of the above.

Q.77) With reference to the types of IPR, which of the following statements is/are incorrect?

1. A patent is granted for an invention which is a new product or process that meets conditions of novelty, non-obviousness and industrial use.
2. An industrial design means a mark capable of being represented graphically and which is capable of distinguishing the goods or services of one undertaking from those of other undertakings.
3. A trademark consists of the creation of a shape, configuration or composition of pattern or color, or combination of pattern and color in three dimensional form containing aesthetic value.

Select the incorrect answer using the code given below:

- a) 1 only
- b) 2 and 3 only
- c) 3 only
- d) 1, 2 and 3

Q.78) Consider the following statements with reference to the Patent pools :

1. Patent pools can be defined as an agreement between two or more patent owners to license one or more of their patents to one another or to third parties.
2. Patent pools may provide an opportunity for a possible anti-competitive behavior.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.79) Consider the following statements with reference to the Patent Law Treaty (PLT):

1. The Patent Law Treaty harmonizes certain patent application procedures in order to reduce or



eliminate formalities and the potential for loss of rights.

2. India is not a signatory to the Patent Law Treaty.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.80) Which of the following statements is/are correct with respect to the IPrism?

- a) It is an Intellectual Property competition for students of schools, polytechnic institutes, colleges and universities.
- b) It aims to foster a culture of innovation and creativity in the younger generation
- c) This competition is for residents of India only.
- d) All of the Above.

Q.81) Paris Convention is for the Protection of -

- a) Patents
- b) Copyright
- c) Trademarks
- d) All of the above.

Q.82) Which of the following statements is incorrect with respect to the National IPR Policy, 2016?

- a) Department of Industrial Policy & promotion (DIPP) is the nodal agency for all IPR issues
- b) The Policy aims to push IPRs as a marketable financial asset, promote innovation & entrepreneurship, while protecting public interest.
- c) The plan will be reviewed every three years in consultation with stakeholders.
- d) The policy is compliant with the WTO's agreement on TRIPS.

Q.83) Which of the following are the items prohibited to export in the flora and fauna category ?

- 1. Sea shells
- 2. Wood and wood products
- 3. Beef of cows, oxen and calves.
- 4. Wood charcoal
- 5. Peacock Tail Feathers & its Handicrafts

Select the correct answer using the code given below :

- a) 1,3,4 and 5 only
- b) 2,4 and 5 only
- c) 1,2,3 and 4 only

d) All the above

Q.84) With reference to the 'Sustainable Sugarcane Initiative (SSI)', consider the following Statements:

- 1. It is an innovative set of agronomic practices that involves using less seeds, raising seeds in a nursery, and following new planting methods.
- 2. It can increase sugarcane yields by at least 20% with 30% less water and a 25% reduction in chemical inputs.
- 3. This method of sugarcane cultivation evolved from the principles of 'More with Less'.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3
- d) 1,2 and 3

Q.85) With reference to 'mRNA Vaccine', consider the following Statements :

- 1. mRNA vaccines use mRNA created in a laboratory to train our cells how to make a protein that will trigger an immune response inside our bodies.
- 2. mRNA vaccines require cell cultures, bacteria or other hosts for growing it

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.86) Recently , 'Convalescent plasma therapy 'has seen news related to -

- a) Treatment for cancer at very low cost
- b) It's a bioremediation technique to clean municipal Waste .
- c) Its passive antibody therapy where transfer of pathogen-specific antibodies to patients.
- d) It's therapy to trigger the immunity of our body .

Q.87) Consider the following Statements :

- 1. Ebola can be transmitted human to human through contact with the blood, secretions, organs and other bodily fluids
- 2. Ebola virus cannot be transmitted From Pregnant women to babies by breast milk, and other pregnancy fluids.



3. Community engagement can be key to successfully control ebola transmission

Which of the statements given above is/are correct about 'Ebola Virus transmission'?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3
- d) 1, 2 and 3

Q .88) With reference to the "'Pradhan Mantri Swasthya Suraksha Nidhi (PMSSN)' consider the following statements:

- 1. It is a non-lapsable reserve fund for Health in the Public Account.
- 2. Administration and maintenance of the PMSSN is entrusted to the Ministry of Finance.

Which of the statements given above is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q .89)"Mission Indradhanush " was seen in news in the context of -

- a) Program to evaluation of values in curriculum to holistic development of children
- b) UNESCO's initiative to incentives gender equality
- c) WHO's mission to vaccinate all countries
- d) It's Immunisation program to children and pregnant women

Q.90) With reference to the ' Nipah Virus' ,consider the following Statements:

- 1. Nipah Virus encephalitis is a zoonotic disease caused by RNA or Ribonucleic acid virus .
- 2. There is no vaccine available and no specific treatment for Nipah Virus.

Which of the Statements given above is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.91) Consider the following diseases:

- 1. Rotavirus
- 2. Smallpox
- 3. Chickenpox
- 4. Yellow fever

Live/ attenuated vaccines are currently available against which of the above viral infections ?

- a) 1, 2, 3 only
- b) 2 and 4 only
- c) 1 and 2, 3 only
- d) All the above

Q.92) With reference to ' The National Nutrition Mission', consider the following Statements:

- 1. The objective of the mission is to improve nutritional outcomes for children, pregnant women and lactating mothers.
- 2. A 100 percent fund will be provided by the center to states and UT for implementation.
- 3. This mission is aided by the World Bank.

Which of the Statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3
- d) 1, 2 and 3

Q.93) With reference to the "National Digital Health Mission", consider the following statements:

- 1. The mission aims to create an integrated healthcare system linking practitioners with the patients digitally.
- 2. The mission can be access and exchange of longitudinal health records of citizens with their consent.
- 3. This mission can create interoperability within the digital health ecosystem.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 1 and 2 only
- c) c) 3 only
- d) 1, 2 and 3 only

Q.94) The 'The Poverty and Shared Prosperity' report is released by -

- a) World Bank
- b) International monetary fund
- c) United Nation Development Program
- d) International Atomic Energy Agency

Q.95) Recently, "The Havana syndrome "mentioned in the news in the context of -

- a) Financial crisis in the European Union due to the Brexit deal.
- b) Psychological condition in which hostages form a bond with their captors or abusers.
- c) Drastically falling oil prices in the USA.



d) It refers to a set of mental health symptoms.

Q.96) With reference to the 'The Food Safety and Standards Authority of India (FSSAI)', consider the following Statements:

1. FSSAI has capped the amount of trans fatty acids (TFA) in oils and fats to 3% for 2021 and 2% by 2022.
2. The WHO has a goal of global elimination of trans fats by 2023.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.97) With reference to 'The Lipoprotein', consider the following Statements:

1. HDL (high-density lipoprotein), also called "bad" cholesterol which poses a high risk for heart disease and stroke.
2. LDL (Low-density lipoprotein), or "good" cholesterol, absorbs cholesterol and carries it back to the liver.

Which of the Statements given above is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.98) With reference to 'Draft Food Safety and Standards (Amendment) Bill, 2020', consider the following Statements:

1. Its aim is to revamp the functioning and jurisdiction of FSSAI.
2. Under this bill the jurisdiction of FSSAI extended to the animal feed industry.
3. FSSAI functions under The Consumer Affairs Ministry.
4. The Food Authority will set up the Food Safety and Standards Authority of India Fund.

Which of the statements given above is/are correct?

- a) 1 2 and 4 only
- b) 2, 3 and 4 only
- c) 1, 3 and 4 only
- d) 1, 2 and 3 only

Q.99) With reference to 'The National Tuberculosis Elimination Program (NTEP)', consider the following Statements:

1. At the national level the program is led by the Central TB Division under the Ministry of Health and Family Welfare.
2. India's goal of eliminating the TB disease is five years ahead of the Sustainable Development Goals target to eliminate TB.
3. "The Moscow Declaration" is related to TB.

Which of the statements given above is/are incorrect?

- a) 1, 2 and 4 only
- b) 2, 3 and 4 only
- c) 1, 3 and 4 only
- d) None the Above

Q.100) With reference to 'The Cancer Genome Atlas (TCGA)', consider the following Statements:

1. It can help the global cancer research community work towards prevention, diagnosis and treatment of the disease.
2. This program was launched by WHO in collaboration with a non-governmental organization, civil society across the world.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2



Q.1) Ans: A

Exp:

- **Gandhisagar dam** is one of the four major dams built on India's **Chambal river**. The dam is located in Mandsaur district of the state of Madhya Pradesh. The Chambal rises near Mhow in the Malwa plateau of Madhya Pradesh and flows northwards through a gorge upwards of Kota. From Kota, it traverses down to Bundi, Sawai Madhopur and Dholpur, and finally joins the Yamuna. The Chambal is famous for its badland topography called the Chambal ravines.
- **Sriram sagar dam** is also known as the **pochampadu project** is the Indian flood- flow project on the **river Godavari**. It has been described as a lifeline for a large part of telangana. The Godavari is the largest Peninsular river system. It is also called the Dakshin Ganga. It rises in the Nasik district of Maharashtra and discharges its water into the Bay of Bengal. The Penganga, the Indravati, the Pranhita, and the Manjra are its principal tributaries.
- **The Idukki dam** is a double curvature arch dam constructed across the **periyar river** in a narrow gorge between two granite hills locally known as kuravan and kurathi in mariyouram village in Idukki district in Kerala, India. The Periyar is the second largest river of Kerala
- **Dantiwada dam** is a mud and masonry dam on the **west Banas river** near dantiwada, Gujarat in India. The dam was constructed in 1965 mainly for irrigation and flood control. Banas rises in the Aravali range and it's a tributary of Chambal.

Reference- <https://bhuvan-app1.nrsc.gov.in/fews/>

Q.2) Ans: B

Exp:

- **Statement 1 is correct** : Bhagat singh was a charismatic Indian revolutionary who participated in the murder of a junior british police officer. He believed in socialist, secular and democratic India. He was inspired by the **Bolshevik revolution**. Being a freedom fighter, he stood for patriotic values and socialism.

- **Statement 2 is correct** : In 1924 he became a member of **Hindustan republican association (HRA)**. Later Bhagat Singh renamed HRA to Hindustan Socialist Republican Association (HSRA).
- **Statement 3 is incorrect** : In 1926, he started a youth organization called **Naujawan Bharat Sabha**. He, along with Batukeshwar Dutt, threw **bombs in the central legislative assembly**. He, along with Rajguru and Sukhdev, was sentenced to death for the Lahore conspiracy Case.
- **Statement 4 is correct** : He wrote an essay **Why I am Atheist**. He furthered the goals of the freedom movement by his revolutionary thinking and courageous acts.

Q.3) Ans: C

Exp:

Green Bonds

- A green bond is a type of debt instrument that proceeds of a green bond offering are earmarked for use towards financing **green projects**. The **World Bank** is a major issuer of green bonds.
- The European green bond standard is a voluntary standard to help scale up and raise the **environmental ambition** of the green bond market.
- Recently, **power finance corporation Ltd (NBFC)** issued a Euro green bond. This bond issuance would help power finance companies in diversifying its currency book as well as the investor base.
- It is the first ever euro denominated green bond issuance from india.
- As of 2021, India has the second largest emerging green bond market after china.
- The most vital characteristic of green bond is that it focuses on garnering positive impact on sustainable development goals and protection of the environment.

Reference-

<https://pib.gov.in/PressReleaselframePage.aspx?PRID=1755380>

Q.4) Ans: A



Exp:

- **Statement 1 is correct**- AUKUS is a security alliance of Australia, USA and UK will focus on advancing strategic interest in the indo-pacific region. Major highlights of this arrangement is the sharing of US nuclear submarine technology to Australia. It will help Australia to acquire **nuclear powered submarines** in cooperation with the UK and the US. This step is significant because the US has only shared nuclear submarine technology once before and it started in 1958 with Great Britain.
- **Statement 2 is incorrect** - After implementation of this arrangement Australia will become the seventh nation in the world to operate nuclear powered submarines after the UK, US, China, Russia, France and India. AUKUS will also involve a new architecture of meetings and engagement between the three countries, as well as cooperation across emerging technologies such as applied AI, quantum technology and under sea capabilities.

Reference:

<https://www.thehindu.com/news/international/explained-will-aukus-upset-the-indo-pacific-balance/article36674358.ece>

Q.5) Ans: B

Exp:

- **Statement 1 is incorrect**- the special drawing right or SDR is neither a currency nor a claim on the IMF. Rather, it is a potential claim on the freely usable currencies of IMF members. The SDR serves as the unit of account of the IMF and some other international organisations.
- **Statement 2 is correct**- the currency value of the SDR is determined by summing the values in US dollars, based on market exchange rates, of a SDR basket of currencies. The SDR currency value is calculated daily and the valuation basket is viewed and adjusted every five years.
- **Statement 3 is incorrect**- the SDR basket of currencies includes the US dollars, Euro, Japanese yen, Pound sterling and Chinese renminbi (included in 2016).
- **Quota** (the amount contributed to the IMF) of a country is denominated in SDR.
 - **Member:** voting power is related directly to the quotas.

- India's foreign exchange reserves also incorporate SDR other than gold reserves, foreign currency assets and reserve trenches in the IMF.

Extra edge by onlyias

International Monetary Fund (IMF)-

- IMF was created in 1945, the IMF is governed by and accountable to the 190 countries that make up its near-global membership. India joined the IMF in december 1945.
- The IMF's primary purpose is to ensure the stability of the international monetary system- the system of exchange rates and international payments that enable countries and their citizens to transact with each other.
- Reports published by IMF-
 1. Global Financial Stability Report
 2. World Economic Outlook

Reference : Onlyias

Q.6) Ans: C

Exp:

- **Statement 1 is correct** : National mission for cultural mapping has now been handed over to Indira Gandhi National center for art, which is gearing up for a trial run of 75 villages in October 2021. **IGNCA** was established in 1987 as an autonomous institution under the **Ministry of culture**, as a centre for research, academic pursuits and dissemination in the field of arts. Cultural ministry was working on the **cultural mapping mission** to build a comprehensive database of artists, arts forms and other resources from organizations under ministry. The work on creating a database for **folk arts and mapping of villages** would be carried out over five years.
- **Statement 2 is correct** : While the entire project would take five years and a Budget of ₹89 crore approved for it, the IGNCA would aim to complete mapping in 5000 villages by the end of financial year 2021-22. Volunteers from the **Nehru Yuva Kendra sangathan, the national service scheme and students of sociology** and social work would be deputed to collect such data from villages.

Extra edge by Onlyias:

Other schemes related to Art and culture

- Kala sanskriti vikas yojana



- Scheme of financial assistance for promotion of art and culture
- Ek Bharat Shreshtha Bharat
- Scheme for safeguarding intangible cultural heritage.

Reference:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1592080>

Q.7) Ans: C

Exp:

About 'SPIN' scheme-

- It is an initiative through which **KVIC** will make it **easier for potters to obtain low-interest loans**, allowing them to diversify their business and increase their income.
- It is aimed at sustainable development by creating local self-employment which is aligned with the Prime Minister's commitment of Job to every hand (har hath me kaam).
- It is a no-subsidy program. KVIC facilitates potters to get back loans under Pradhan Mantri Shishu Mudra Yojana. No financial burden on the exchequer. Beneficiaries can repay the loan in easy installments.
- **Significant-** It will reduce their dependence on government subsidy and thus make potters self-reliant.

Extra edge by Onlyias-

Khadi and village industries commission (KVIC)-

- A statutory body established under the khadi and village industries commission act, 1956.
- It functions under the Ministry of Micro, Small and Enterprises.
- In 2017-18, it launched the Honey Mission programme.
- Recently, it launched a project named Bamboo Oasis on Lands in Drought (BOLD) from the village nichkamandwa in Udaipur Rajasthan.

Q.8) Ans: B

Exp:

About Resolution 2593

- The resolution called on the **Taliban** to keep their commitments on preventing terror groups in afghanistan.

- It urged them to assist the safe evacuations of all afghan nationals wishing to lead the countries.
- It mentions individuals designated by Resolution 1267 also, which includes the Laskar E Taiba and Jaise-e-Muhammad.

Extra edge by Onlyias

UNSC-

- The UNSC is one of the six principal organs of the United Nations.
- Its purpose is to be charged with maintenance of international peace and security.
- Powers- establishment of peacekeeping operations, establishment of international sanctions, and the authorization of military action through Security Council resolution.
- It is the only UN body with the authority to issue binding resolutions to member states.
- Members- the security consists of fifteen members. Permanent members (P5): Russia, UK, France, China and USA (have veto power) and 10 non permanent members, elected on a regional basis to serve two-year terms.
- The body's Presidency rotates monthly among its 15 members.

Reference:

<https://www.thehindu.com/news/national/uns-resolution-on-taliban-serves-indias-interest/article36198479.ece>

Q.9) Ans: A

Exp:

- **Statement 1 is correct:** An Account Aggregator (AA) is a type of RBI regulated entity that helps an individual securely and digitally access and share information from one financial institution they have an account with to any other regulated financial institution in the AA network. Data cannot be shared without the consent of the individual. The Account Aggregator system can make lending and wealth management a lot faster and cheaper.
- **Statement 2 is incorrect:** The individual's bank just needs to join the Account Aggregator network. India's financial system involves many hassles for customers today -- The Account Aggregator network would replace traditional methods with a simple, mobile-based, simple, and safe digital data access & sharing process. Registering with an AA is fully voluntary for customers. A customer can



reject the consent to share a request at any time.

The two key services that will be improved for an individual is access to loans and access money management.

Source : <https://www.livemint.com/money/personal-finance/account-aggregators-these-banks-have-joined-it-how-it-will-benefit-customers-11631417562966.html>

Q.10) Ans: B

Exp:

- **Statement 1 is incorrect:** World Heritage Institute of Training and Research –Asia Pacific is a non-profit organization specialized in the area of heritage conservation. It is an Institute under the auspices of UNESCO, it is the first one established in developing countries. It is an autonomous institution at the service of member states and associate members of UNESCO
- **Statement 2 is correct:** The Mission of the institution is to strengthen implementation of the World Heritage Convention 1972 in the Asia and the Pacific region. Recently it has recognised CEPT University's Programme in Conservation and Regeneration as a "commended case" for its focus on sustainability, people-centered approach, and innovative reforms in heritage conservation.

Source:

<https://indianexpress.com/article/cities/ahmedabad/world-heritage-education-recognises-cept-varsity-programme-7481226/>

Q.11) Ans: C

Exp:

- **Statement 1 is correct:** The Ubreathe Life is the world's first, state-of-the-art 'Smart Bio-Filter' that can purify the air. It amplifies the air purification process in indoor spaces. These indoor spaces can either be hospitals, schools, offices and your homes. IIT Ropar, which is a designated iHub – AWaDH (Agriculture and Water Technology Development Hub) of the Department of Science and Technology has developed the Ubreathe Life.
- **Statement 2 is correct:** The technology works through the air-purifying natural leafy plant. The room air interacts with leaves and goes to the soil-root zone where maximum pollutants are purified. The technology used in this product is the 'Urban Munnar Effect' along with patent-pending "Breathing Roots" to exponentially amplify the

phytoremediation process of the plants. Ubreathe Life' effectively improves indoor air quality by removing particulate, gaseous, and biological contaminants while increasing the oxygen levels in the indoor spaces.

Source:

<https://timesofindia.indiatimes.com/home/education/news/iit-ropars-startup-company-introduces-worlds-first-plant-based-smart-air-purifier-ubreathe-life/articleshow/85833943.cms>

Q.12) Ans: D

Exp:

International Fund for Agricultural Development

- The International Fund for Agricultural Development (IFAD) is a specialized international financial institution of the United Nations working in the field of poverty eradication in the rural areas of developing countries.
- It was created in 1977, which is the outcome of the World Food Conference of 1974. It invests in rural people, empowering them to increase their food security, improve the nutrition of their families and increase their incomes.
- It helps developing countries to build resilience, expand their businesses and take charge of their own development.
- It's Headquarter is situated in Rome, Italy. It has 177 member countries and India is also a member country.

Source: <https://www.ifad.org/en/>

Q.13) Ans: B

Exp:

- **Statement 1 is incorrect:** Articles 124(2) of the Constitution govern the appointment of judges to the Supreme Court (SC). Under this, the President has the power to make the appointment after "consultation" with such judges of SC as the president may deem necessary.
- **Statement 2 is correct:** The SC collegium is headed by the CJI and comprises four other senior most judges of the court. Judges of the higher judiciary are appointed only through the collegium system and the government has a role only after names have been decided by the collegium.

Source

<https://indianexpress.com/article/explained/explaine>

d-how-supreme-court-judges-are-appointed-7487796/

Q.14) Ans: C

Exp:

- **Statement 1 is correct:** Nipah Virus is a zoonotic virus as it is transmitted from animals to humans. The first outbreaks of the Nipah virus among humans was reported from Malaysia (1998) and Singapore (1999). The animal host reservoir for this virus is known to be the fruit bat, commonly known as flying fox.
- **Statement 2 is correct:** Fruit bats are known to transmit this virus to other animals like pigs, and also dogs, cats, goats, horses and sheep. The transmission happens mainly through direct contact with these animals or through the consumption of food contaminated by the saliva or urine of these infected animals. Person-to-person transmission is not fully established yet. Currently, there are no vaccines for both humans and animals. Intensive supportive care is given to humans infected by Nipah virus.

Source : <https://www.cbsnews.com/news/nipah-virus-outbreak-india-kerala/>

Q.15) Ans: B

Exp:

- Animal spirits' is a term coined by the British economist, J. M. Keynes. This term refers to the ways that human emotions and instincts can drive financial decision-making of investors and consumers in times of economic stress or uncertain environments and volatile times.
- Animal spirits are used to help explain why people behave irrationally, and are the forerunner to modern behavioral economics. If spirits are low, confidence levels will be low. This will drive down a promising market even if the economy fundamentals are strong. If spirits are high, confidence among participants in the economy will be high, and market prices will be high.

Source: <https://www.merriam-webster.com/dictionary/animal%20spirits>

Q.16) Ans: B

Exp:

- **V. O. Chidambaram Pillai was** Born on September 5, 1872, Pillai is popularly known as 'Kappalottiya Tamizhan' or "The Tamil Helmsman.

- He is best remembered for the venture he ran against tremendous odds, the Swadeshi Steam Navigation Company, to challenge the monopoly of the British in sea trade.
- He launched the first indigenous Indian shipping service between Tuticorin and Colombo with the Swadeshi Steam Navigation Company (SSNC), competing against British ships.
- Pillai's ambitions were high—though services were restricted to Tuticorin and Colombo, Swadeshi aimed to "popularize the art of Navigation" among "Nations of the East".

Source:

<https://pib.gov.in/PressReleaseFramePage.aspx?PRID=1752201>

Q.17) Ans: A

Exp:

- **Statement 1 is correct:** Bitcoin is a consensus network that enables a new payment system and a completely digital money. It is the first decentralized peer-to-peer payment network that is powered by its users with no central authority or middlemen. From a user perspective, Bitcoin is pretty much like cash for the Internet. Bitcoin can also be seen as the most prominent triple entry bookkeeping system in existence.
- **Statement 2 is incorrect:** El Salvador became the first country in the world to adopt bitcoin as legal tender. It traditionally was an agricultural country, heavily dependent upon coffee exports.

Source : <https://bitcoin.org/en/faq>

Q.18) Ans: C

Exp:

- **Statement 1 is correct:** Common Services Centre (CSC) programme is an initiative of the Ministry of Electronics & IT (MeitY), Government of India. CSCs are the access points for delivery of various electronic services to villages in India, thereby contributing to a digitally and financially inclusive society. CSCs are more than service delivery points in rural India.
- **Statement 2 is correct:** CSC e-Governance Services India Limited is a Special Purpose Vehicle (CSC SPV) incorporated under the Companies Act, 1956 by the Ministry of Electronics and Information Technology (MeitY), Government of India, to monitor the implementation of the Common Services Centers Scheme. It provides a centralized



collaborative framework for delivery of services to citizens through CSCs, besides ensuring systemic viability and sustainability of the scheme.

Source: <https://vikaspedia.in/e-governance/resources-for-vles/common-service-centres-programme>

Q.19) Ans: D

Exp:

Fragile X Syndrome

- Fragile X Syndrome is a genetic disorder that causes a range of developmental problems including learning disabilities and cognitive impairment.
- It is the leading inherited cause of autism in 4% of the global population. It is caused by changes in FMR1 gene which make an important protein (FMRP) that is required for brain development.
- It is caused by the mutation named FRAXA. Males are more severely affected by this disorder than females. The simplest tool for timely detection is a DNA test. This testing could be made mandatory for every child diagnosed with autism.
- Symptoms are Learning difficulty, speech delay, aggressive behaviour, hyperactivity, attention deficit, fear of the unfamiliar, sensory processing disorders and problems in motor skills.

Source:

<https://www.nichd.nih.gov/health/topics/fragilex>

Q.20) Ans: B

Exp:

About raja Mahendra Pratap Singh-

- Raja Mahendra Pratap Singh was a freedom fighter, social reformer and a figure representing the Jat community, predominant in western Uttar Pradesh.
- He was well versed in **8 languages** and practiced different religions.
- He established the provisional government of India at Bagh-e-Babur in Kabul in the middle of world war I 1915. He declared himself president and his fiery fellow revolutionary Maulana barakatullah of Bhopal, prime minister of the provincial government.
- In 1929, he launched the **world federation in Berlin**. Moreover he was nominated for the Nobel peace prize in 1937.

- He returned to India a year before Independence and immediately began work with Mahatma Gandhi. In free India, he diligently pursued his ideal of Panchayati Raj. He won Lok Sabha elections as an Independent candidate from Mathura in 1957.

Reference:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1754431>

Q.21) Ans: B

Exp:

Temperate deciduous forest

- Temperate deciduous or temperate broad-leaf forests are a variety of temperate forest 'dominated' by trees that lose their leaves each year. They are found in areas with warm moist summers and cool winters.
- A **temperate forest** is a forest found between the tropical and boreal regions, located in the temperate zone. It is the second largest biome on the planet, covering 25% of the world's forest area, only behind the boreal forest, which covers about 33%.
- These forests cover both hemispheres at latitudes ranging from 25 to 50 degrees, wrapping the planet in a belt similar to that of the boreal forest. Due to its large size spanning several continents, there are several main types: deciduous, coniferous, mixed forest, and rain forest.
- The climate of a temperate forest is highly variable depending on the location of the forest. For example, Los Angeles and Vancouver, Canada are both considered to be located in a temperate zone, however, Vancouver is located in a temperate rainforest, while Los Angeles is more subtropical.

Extra edge by only IAS

Tropical rainforest

- Tropical rainforests are rainforests that occur in areas of tropical rainforest climate in which there is no dry season. All months have an average precipitation of at least 60 mm – and may also be referred to as *lowland equatorial evergreen rainforest*. True rainforests are typically found between 10 degrees north and south of the equator, they are a subset of the tropical forest biome that occurs roughly within the 28-degree latitudes (in the equatorial zone between the tropic of cancer and tropic of capricorn).

Dry deciduous forest

- Tropical dry deciduous forests can be found in severe and extremely variable climates characterized by low annual rainfall, 5-6 dry months within the annual cycle, and nutrient-poor soil. Several terms have been used for this vegetation type such as seasonally dry tropical forest, tropical dry deciduous forest, monsoon forest, caatinga, cuabal, etc.

Coniferous forest

- The coniferous forest is sandwiched in between the tundra to the north and the deciduous forest to the south. One type of coniferous forest, the northern boreal forest, is found in 50° to 60°N latitudes. Another type, temperate coniferous forests, grows in lower latitudes of North America, Europe, and Asia, in the high elevations of mountains.

- India's Mahatma Gandhi National Rural Employment Guarantee Scheme (MGNREGS) is the world's largest public works-based social protection programme, and its dual focus on wages and natural resource management assets provides the opportunity to enhance the climate resilience outcomes of the programme.
- Evidence shows that MGNREGS is already helping communities cope with climate impacts to some extent. With the inclusion of climate information services in planning, implementation and monitoring, the scheme has the potential to help communities undertake climate risk management more effectively.
- The tool has already been successfully piloted in 18 village panchayats of Niwali Block in Badwani, Madhya Pradesh, and CRISP-M shows the value of combining scientific climate risk information, methods, practices and technology with the traditional knowledge, practices and skills of communities.

Source: <https://www.iied.org/launch-crisp-m-tool>

Q.22) Ans: D

Exp:

- Statement 1 is Incorrect:** Vidyanjali is an initiative taken by the **Ministry of Education**, Government of India. It aims to strengthen Schools through community and private sector involvement in schools across the country. This initiative would connect schools with varied volunteers from the **Indian Diaspora namely, young professionals, school alumni, in service and retired teachers / Government officials / professionals** and many others.
- Statement 2 is Incorrect:** Vidyanjali has **two** verticals in which volunteers can support and strengthen the **government** and **government aided** schools.
 1. **Participate in school Service/Activity.**
 2. **Assets/Material/Equipment.**

Reference:

<https://vidyanjali.education.gov.in/en/about-us>

Q.23) Ans: A

Exp.

- A new digital tool that combines scientific climate risk information with the traditional knowledge, practices and skills of communities to help Indian communities undertake climate risk management more effectively.

Q.24) Ans: B

Exp:

- Statement 1 is incorrect:** Kalamkari means creating something with a 'kalam', or a brush. Traditionally, only three colours are used — black, red and yellow. But lately, 'pale blue' has also started being used. The art forms are traditional in nature. The paintings are intricately made on a cotton cloth using pens or brushes made out of bamboo tree and coconut tree stems. This tradition is famous in Tamil Nadu.
- Statement 2 is correct:** The Karuppur kalamkari paintings recently received the geographical indication (GI) tags. GI is a tag used on products with a specific geographical origin, and ones that possess a reputation due to that origin.

Sources:

<https://indianexpress.com/article/lifestyle/art-and-culture/karuppur-kalamkari-paintings-kallakurichi-wood-carvings-get-the-geographical-indication-gi-tags-7568428/>

Q.25) Ans: C

Exp:

- Statement 1 is correct:** The Kunming Declaration was adopted by over 100 countries at the first part

of the ongoing virtual 15th meeting of the Conference of the Parties to the United Nations Convention on Biological Diversity. It calls upon the parties to “mainstream” biodiversity protection in decision-making and recognise the importance of conservation in protecting human health.

- **Statement 2 is correct:** By adopting this declaration the nations have committed themselves to support the development, adoption and implementation of an effective post-2020 implementation plan, capacity building action plan for the Cartagena Protocol on biosafety. The theme of the declaration is Ecological Civilization: Building a Shared Future for All Life on Earth.

The declaration stated:

- Putting biodiversity on a path to recovery is a defining challenge of this decade, in the context of the UN Decade of Action for Sustainable Development, the UN Decade on Ecosystem Restoration and the UN Decade for Ocean Science for Sustainable Development.
- It requires strong political momentum to develop, adopt and implement an ambitious and transformative post-2020 global biodiversity framework that promotes the three objectives of the Convention in a balanced manner.
- The parties have demonstrated their commitment to accelerate development and update of the strategy document National Biodiversity Strategies and Action Plans.

Source:

<https://www.downtoearth.org.in/news/wildlife-biodiversity/over-100-countries-sign-kunming-declaration-on-biodiversity-conservation-79685>

Q.26) Ans: B

Exp:

- **Statement 1 is incorrect-** the FSDC is a non-statutory apex council under the Minister of Finance constituted by the executive order in 2010. The Raghuram Rajan committee (2008) on financial sector reform first proposed the creation of FSDC.
- Functions of FSDC is to strengthen and institutionalize the mechanism for maintaining financial stability, enhancing inter-regulatory coordination and promoting financial sector development.
- **Statement 2 is correct-** Account aggregator is a non banking financial company engaged in the business of providing the service of retrieving or

collecting financial information pertaining to its customer, under a contract. It was created through an inter-regulatory decision by RBI and other regulators including SEBI, IRDA and PFRDA through an initiative of the Financial Stability and development Council (FSDC).

- The objective of Account Aggregator is to aggregate all financial information of an individual.
- **Statement 3 is correct-** the chairperson of the FSDC is the Union Finance Minister of India. And the FSDC subcommittee is chaired by the Governor of RBI. members of FSDC are Governor RBI, Finance secretary and chairman of SEBI, IRDA, PFRDA and IBBI. All the members of the FSDC are also members of the sub-committee.

Q.27) Ans: A

Exp:

UN Environment programme

- The United Nations environment programme (UNEP) is the leading global environmental authority that sets the global environmental agenda, promotes the coherent implementation of the environmental dimension of sustainable development within the United Nations system, and serves as an authoritative advocate for the global environment.
- Headquartered in Nairobi, Kenya. It was founded as a result of the United Nations conference on the human environment 1972.
- The UNEP is headed by an executive director.
- It assists developing countries in implementing environmentally sound policies and practices. It has overall responsibility for environmental problems among United Nations agencies.
- The UNEP has a few focus areas in which they prioritize their work such as climate change, ecosystem management, disaster and conflict, environmental governance, resource efficiency, chemical and waste etc.
- Major reports published by UNEP:
 - **Emissions gap report.**
 - **Global environment outlook**
 - **Frontiers**
 - **invest into healthy planet**
- Major campaign: Beat pollution, UN75, world environment day, wild for life, billion tree campaign.

Q.28) Ans: C

Exp:

- **Statement 1 is correct :** CBD is a legally binding convention recognised for the first time, that the conservation of biological diversity is a 'common concern of humankind' and is an integral part of the development process. The agreement covers all ecosystem, species and genetic resources. The objective of CBD is: the sustainable use of its components and the fair and equitable sharing of the benefits arising out of the utilisation of genetic resources, including by appropriate access to genetic resources and by appropriate transfer of relevant technologies, taking into account all rights over those resources and to technologies, and by appropriate funding.
- **Statement 2 is correct :** The convention of conservation of migratory species of wild animals (also known as Bonn convention) aims to conserve terrestrial, aquatic and avian migratory species throughout their range. It is an intergovernmental treaty, concluded under the aegis of the United Nations environment programme, concerned with the conservation of wildlife and habitats on a global scale.
- **Statement 3 is incorrect :** Created in 1988 by the World Meteorological Organization (WMO) and the United Nations Environment Programme (UNEP), the objective of the IPCC is to provide governments at all levels with scientific information that they can use to develop climate policies. The IPCC currently has 195 members including India.
- **Statement 4 is correct :** TRAFFIC is a joint conservation programme of WWF and IUCN. It was established in 1976 by the species survival commission of IUCN, principally as a response to the entry into force during the previous years of the convention on the International Trade in Endangered Species of Wild fauna and Flora (CITES). TRAFFIC is an international network, consisting of TRAFFIC international based in Cambridge UK with offices on five continents.

Q.29) Ans: B

Exp:

- **Statement 1 is incorrect-** The animal welfare board of India is a statutory advisory body on animal welfare laws and promotes animal welfare in the country. The animal welfare board of India, the first of its kind to be established by any Government in the world, was set up in 1962, in

accordance with section 4 of the prevention of cruelty to animals act, 1960.

- Shrimati Rukmani Devi arundale, pioneered the setting up of the board, with its headquarters at Chennai. She guided the activities of the board for nearly twenty years till her demise in 1986.
- **Major functions:** to keep the law in force in India for the prevention of cruelty to animals under constant study and to advise the government on the amendments to be undertaken in any such law from time to time.
- To advise the government or any local authority or other person on improvements in the design of vehicles so as to lessen the burden on draught animals.
- To advise the central government on the making of rules under the act with a view to preventing unnecessary pain or suffering to animals generally, and more particularly when they are being transported from one place to another or when they are used as performing animals or when they are kept in captivity or confinement.
- **Statement 2 is correct-** its other function is to give financial assistance and other assistance to Animal Welfare Organisations, functioning in any local area or to encourage the formation of Animal Welfare organizations in any local area which shall work under the general supervision and guidance of the board.
- The board consists of 28 members. The term of office of members is for a period of 3 years.

Q.30) Ans: B

Exp:

- **About Nilgiri tahr-** Nilgiri tahr is also known as nilgiri ibex. It has been listed as 'endangered' by IUCN. It has been listed under schedule 1 of the wildlife protection act, 1972. It is found in open montane grassland habitat of rain forests ecoregion. It is endemic (only found in) to the Nilgiri hills and the southern portion of the western ghats in Tamilnadu and Kerala, India. It is the state animal of Tamil Nadu.



- **Red panda**- the red panda is a small reddish-brown arboreal mammal. The only living member of the genus *Ailurus*. It is also the state animal of Sikkim. It is listed as endangered in the IUCN red list of threatened species and under schedule 1 of the Indian wildlife (protection) act, 1972. It is found in the forests of India, Nepal, Bhutan and the northern mountains of Myanmar and southern China.



- **Kashmir stag**- also known as Hangul, is the state animal of Jammu and Kashmir, is restricted to the Dachigam national park. The hangul is placed under schedule 1 of the Indian wildlife protection act, 1972. The hangul was once widely distributed in the mountains of Kashmir and parts of Chamba district in neighbouring Himachal Pradesh. It is listed as critically endangered by IUCN.



- **Indian buffalo**- the Indian buffalo is mainly found in the alluvial grasslands, marshes, swamp and river valleys. They are generally found in areas that have plenty of water holes and resources. It is listed under schedule 1 of the wildlife protection act, 1972. It is classified as endangered in the IUCN red list of threatened species.



Exp:

- **Statement 1 is correct:** It lives at high altitudes in the steep mountains of Central and Southern Asia, where the climate is extremely cold. The Snow Leopard (also known as Ghost of the mountains) acts as an indicator of the health of the mountain ecosystem in which they live. It is because of their position as the top predator in the food web.
- **Statement 2 is correct:** It is also the State animal of Himachal Pradesh. Further, it is found in 12 range countries namely Afghanistan, Bhutan, China, India, Kazakhstan, Kyrgyzstan, Mongolia, Nepal, Pakistan, Russia, Tajikistan, and Uzbekistan. In India, it inhabits the higher Himalayan and trans-Himalayan landscape in the states/union territories of Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Sikkim, and Arunachal Pradesh.

Conservation Status:

- IUCN Status: Vulnerable
- CITES: Appendix I
- Indian Wildlife (Protection) Act 1972: Schedule I

Source : <https://www.firstpost.com/india/ladakh-announces-snow-leopard-as-state-animal-black-necked-crane-as-state-bird-9930411.html>

Q.32) Ans: A

Exp:

Phytoremediation

- Phytoremediation technologies use living plants to clean up soil, air, and water contaminated with hazardous contaminants. It is defined as "the use of green plants and the associated microorganisms, along with proper soil amendments and agronomic techniques to either contain, remove or render toxic environmental contaminants harmless".
- It is proposed as a cost-effective plant-based approach of environmental remediation that takes advantage of the ability of plants to concentrate elements and compounds from the environment and to detoxify various compounds. The concentrating effect results from the ability of certain plants called hyperaccumulators to bioaccumulate chemicals.
- Toxic heavy metals cannot be degraded, but organic pollutants can be and are generally the major targets for phytoremediation.

Source:

<https://en.wikipedia.org/wiki/Phytoremediation>

Q.31) Ans: C



Q.33) Ans: D

Exp:

- **Pair 1 is correct** : The Carpathian Convention, Framework Convention on the Protection and Sustainable Development of the Carpathians is a framework type convention pursuing a comprehensive policy and cooperating in the protection and sustainable development of the Carpathians. It is joint policies for sustainable development among the **seven countries of the region** Czech Republic, Hungary, Poland, Romania, Serbia and Montenegro, Slovakia and Ukraine.
- **Pair 2 is correct** : The Bamako Convention, Bamako Convention on the Ban on the Import into Africa and the Control of Transboundary Movement and Management of Hazardous Wastes within Africa is a **treaty of African nations prohibiting the import of any hazardous (including radioactive) waste**. The convention was negotiated by twelve nations of the Organisation of African Unity at Bamako, Mali in January, 1991, and came into force in 1998.
- **Pair 3 is correct** : The Tehran Convention, Protection of the Marine Environment of the Caspian Sea. It serves as a governance framework which lays down the general requirements and the institutional mechanism for environmental protection and sustainable development in the Caspian Sea region.

Source : Wikipedia

Q.34) Ans: B

Exp:

- The CCPI is developed by **not-for-profit organisations Germanwatch and NewClimate Institute (Germany) together with the Climate Action Network (CAN International)** annually since 2005.
- It is an important tool to enhance transparency in international climate politics and enables comparison of climate protection efforts and progress made by individual countries.
- It is an independent monitoring tool for tracking the climate protection performance of 57 countries and the European Union.
- The CCPI considers four categories, with 14 indicators:
 - Greenhouse Gas Emissions (40% of the overall score)
 - Renewable Energy (20%)

- Energy Use (20%)
- Climate Policy (20%).

- India ranked 10th and scored 63.98 points out of 100.
- It showcases only two G20 countries the UK and India are among the high rankers while six others the USA, Saudi Arabia, Canada, Australia, South Korea and Russia are at the bottom of the index.

Source: https://m.timesofindia.com/india/india-in-top-10-performers-on-climate-change-index/amp_articleshow/79612075.cms

Q.35) Ans: B

Exp:

- **The Ministry of Earth Sciences (MoES)** has recently published a Climate Change report entitled "Assessment of Climate Change over the Indian Region" which covers all the aspects of regional climate change including the climatic extremes across India.
- The preparation of this report was led by the Center for Climate Change Research (CCCR) at the Indian Institute of Tropical Meteorology (IITM) Pune. The report from the MoES is the first of its kind where a comprehensive discussion has been made regarding the impact of human-induced global climate change on the regional climate and monsoon of the Indian subcontinent, adjoining the Indian Ocean and the Himalayas.
- **Highlights of report**
 - Hindu Kush Himalayas experienced a temperature rise of 1.3°C during 1951–2014. Projected rise by the end of the 21st century is 5.2°C.
 - Sea surface temperature of the tropical Indian Ocean rose by 1°C on average during 1951–2015 (greater than global average SST warming 0.7°C).
 - Droughts, both the frequency and spatial extent, have increased significantly during 1951–2016

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1706938>

Q.36) Ans: D

Exp:

Seaweeds

- **Seaweeds** are (thalloid plants) macroscopic algae, which means they have **no differentiation of true tissues such as roots, stems and leaves**. They have leaf-like appendages.
- Seaweeds, the larger and visible marine plants, are found attached to rocks, corals and other submerged strata in the intertidal and shallow sub tidal zones of the sea.
- Seaweeds grow in **shallow coastal waters wherever a sizable substrata is available**.
- Based on the colour of their pigmentation, seaweeds are broadly classified into different classes such as
 - Blue- green
 - Green
 - Brown
 - Red etc.

Functions of seaweeds :

- Food for marine organisms
- Habitat for fish breeding grounds.
- Source of sediment.

Uses of seaweeds :

- Seaweeds are important as food for humans, feed for animals, and fertilizer for plants.
- **Seaweeds are used as a drug for goiter treatment, intestinal and stomach disorders.**
- Products like agar-agar and alginates, iodine which are of commercial value, are extracted from seaweeds.

Source: Environment Book by Shankar IAS ,CHAPTER - 14 MARINE ORGANISMS

Q.37) Ans: A

Exp:

- **Statements 1,2 and 4 are correct:** Sacred groves comprise patches of forests or natural vegetation from a few trees to forests of several acres that are usually dedicated to local folk deities.
- These spaces are protected by local communities because of their religious beliefs and traditional rituals that run through several generations.
- **Ecological Significance of sacred groves :**
- **Conservation of Biodiversity** – The sacred groves are important repositories of floral and faunal diversity that have been conserved by local

communities in a sustainable manner. They are often the last refuge of endemic species in the geographical region.

- **Recharge of aquifers** – The groves are often associated with ponds, streams or springs, which help meet the water requirements of the local people. The vegetative cover also helps in recharging the aquifers.
- **Soil conservation** - The vegetation cover of the sacred groves improves the soil stability of the area and also prevents soil erosion.

Extra Edge by OnlyIAS

Classification of Sacred Groves :

- **Traditional Sacred Groves** : It is the place where the village deity resides, who is represented by an elementary symbol.
- **Temple Groves** : Here a grove is created around a temple and conserved.
- **Groves around the burial or cremation grounds.**

Distribution of Sacred Groves in India :

- In India, sacred groves are found all over the country and abundantly along the western ghats in the states of Kerala and Karnataka.

Source : Environment Book by Shankar IAS ,CHAPTER - 15 PROTECTED AREA NETWORK

Q.38) Ans: A

Exp:

- **Statement 1 is correct:** It is also known as Asiatic wild dog, red dog, dog. It is about the size of a German shepherd, but looks more like a long legged fox. Furthermore, it is a highly social animal, living large in clans without rigid dominance hierarchies and containing multiple breeding females. They are native to Central , South , East Asia, and Southeast. Dholes play an important role in Asia. role as apex predators in forest ecosystems .
- **Statement 2 is incorrect:** Dholes occupy a wide variety of climates and habitats, including dense forests, scrub, steppes, and alpine regions. They vary in colour from charcoal grey to rust-red to sandy beige , depending on their habitat. In India they are found in Western and Eastern Ghats, Central North East India.

Conservation Status:



- **Schedule II** species under the Wildlife Protection Act, 1972.
- **Endangered** by the International Union for Conservation (IUCN)
- **Under Appendix II** of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) list.

<https://www.downtoearth.org.in/news/wildlife-biodiversity/conserving-india-s-dhole-population-study-earmarks-114-talukas-79636>

Q.39) Ans: A

Exp:

- **Statement 1 is correct:** Vision Plan (2021-2031) for Indian Zoos was recently released. The vision plan is aimed at upgrading Indian zoos and strengthening the Central Zoo Authority (CZA). The CZA is a statutory body under the Ministry of Environment, Forest and Climate Change. It was constituted in 1992 under the Wildlife (Protection) Act, 1972.
- **Statement 2 is incorrect:** It is chaired by the Environment Minister and has 10 members and a member secretary. The main objective of the authority is to complement and strengthen the national effort in conservation of rich biodiversity.

Important functions are:

- It enforces minimum standards and norms for the upkeep and healthcare of animals in Indian zoos.
- Every zoo in the country is required to obtain recognition of its operation and It can also de-recognise zoos.
- It coordinates and implements programmes on capacity building of zoo personnel, planned breeding programmes and ex-situ research.

Source:

<https://www.latestly.com/socially/india/news/vision-plan-for-indian-zoos-released-focus-on-local-fauna-conservationread-here-latest-tweet-by-prasar-bharati-news-services-2944130.html>

Q.40) Ans: D

Exp:

- **Stubble burning** is a common practice followed by farmers to prepare fields for sowing wheat in November as there is little time left between the harvesting of paddy and sowing of wheat. Stubble burning results in emissions of harmful gases such as carbon dioxide, sulphur dioxide, nitrogen dioxide along with particulate matter.

- Open stubble burning emits large amounts of toxic pollutants in the atmosphere which contain harmful gases like methane (ch₄), carbon monoxide (co), volatile organic compounds and carcinogenic polycyclic aromatic hydrocarbons. They may eventually cause smog.
- **Alternative solution that can avoid stubble burning-**
 - Promote paddy Straw based plants.
 - Incorporation of crop residue in the soil.
 - Convert the removed residues into enriched organic manure.
 - Extraction of yeast proteins can be explored through scientific approach.

Extra edge by Onlyias

Chhattisgarh model-

An innovative experiment has been undertaken by chhattisgarh government by setting up gauthans

- A gauthan is a dedicated five acre plot held in common by each village where all the usable parali is collected through parali daan and is converted into organic fertilizer by mixing with cow dung and few natural enzymes.
- The scheme also generates employment among rural youth.
- The government supports the transportation of parali from the farm to the nearest gauthans.

Refernce-

<https://pib.gov.in/PressReleaseDetail.aspx?PRID=1707021>

Q.41) Ans: B

Exp:

- **Statement a is incorrect :** Biomedical waste comprises human and animal anatomical waste, treatment apparatus like needles, syringes and other materials used in health care facilities in the process of treatment and research. This waste is generated during diagnosis, treatment or immunisation in hospitals, nursing homes, pathological laboratories, blood banks etc.
- **Statement b is correct :** The hospitals servicing 1000 patients or more per month are required to obtain authorisation and segregate biomedical waste into 10 categories, pack five colour backs for disposal.
- **Statement c is incorrect :** The quantum of waste generated in India is estimated to be 1-2 kg per bed per day in a hospital and 600 gm per day in a clinic.



85% of the hospital waste is non- hazardous, 15% is infections/hazardous. Mixing of hazardous materials results in contamination and makes the entire waste hazardous. Hence, there is a necessity to segregate and treat. Improper disposal increases risk of infection, encourages recycling of prohibited disposables and disposed drugs and develops resistant microorganisms.

- Scientific disposal of biomedical waste through segregation, collection and treatment in an environmentally sound manner minimises the adverse impact on health workers and on the environment.
- Total bio-medical waste generation in the country is 484 TDP from 1,68,869 health facilities (HCF), out of which 447 TDP is treated.

Reference-

<https://pib.gov.in/Pressreleaseshare.aspx?PRID=1526326>

Q.42) Ans: B

Exp:

- **Statement 1 is correct** : Faster adoption and manufacturing of hybrid and electric (FAME india) scheme rolled out in April, 2015 by the Union Minister of heavy industries and public enterprises. The scheme will help promote use of electric and hybrid vehicles and initially, a subsidy of 30% will help be provided to the buyers. Phased replacement of fossils on latest technology will lead to a net saving of Rs 14000 crore. The scheme is proposed to be implemented over six year, till 2020 which looks at sales of electric and hybrid vehicles up to 60-70 lakh units per year.
- **Statement 2 is incorrect** : Carbon sequestration and storage also, known as CCS and carbon sequestration, describes the technologies designed to tackle global warming by capturing CO₂ at power stations, industrial sites or even directly from the air and permanently storing it underground. Carbon sequestration describes long-term storage of carbon dioxide or other forms of carbon to either mitigate or defer global warming. It has been proposed as a way to slow the atmospheric and marine accumulation of greenhouse gases, which are released by burning fossil fuel. Two type or sink

Natural sink- ocean, forest, soil etc.

Artificial sink- depleted oil reserves, unminerable mines etc.

- Types of sequestration

- hydrodynamic trapping- carbon dioxide can be trapped as a gas under low- permeability cap rock. Natural gas is stored in gas reservoirs.
- solubility trapping
- mineral carbonation- Carbon dioxide can be effectively stored in the earth's subsurface by hydrodynamic trapping and solubility trapping - using a combination of the two is most effective.

Reference-

<https://pib.gov.in/PressReleasePage.aspx?PRID=1604991>

Q.43) Ans: A

Exp:

- An eco-friendly product, which is a complete substitute of the plastic in all uses has not been found till date. In the absence of a substitute alternative, it is impractical and undesirable to impose a blanket ban on the use of plastic all over the country. The real challenge is to improve plastic waste management systems.
- **Plastic waste management rules 2021- Ministry of environment forest and climate change** has notified the plastic waste management rules 2021. The rules prohibit specific single use plastic items which have low utility and high litteral potential.
- **Statement 1 is correct- According to new rules-** the manufacture, import, stocking distribution sale and use of the identified single use plastic will be prohibited with effect from 1st July 2022. The ban will not apply to commodities made of compostable plastic.
- For banning other plastic commodities in the future, other than those that have been listed in this notification the government has given industry ten years from the date of notification for compliance.
- **Statement 2 is incorrect-** the permitted thickness of the plastic bags, currently 50 microns will be increased to 75 microns from 30th September, 2021 and to 120 microns from the 31st December, 2022. Plastic bags with higher thickness are more easily handled as waste and have higher recyclability.
- **Legal framework for banning plastic:** currently, the plastic waste management rules, 2016 prohibits manufacture, import, distribution sale and use of canary bags and plastic sheets less than 50 microns in thickness in the country.



- **Implementing agency:** the Central pollution Control board, along with the state pollution board will monitor the ban, identify violations and impose penalties already prescribed under the Environmental protection act, 1986.

Reference-

<https://pib.gov.in/PressReleaseframePage.aspx?PRID=1745433>

Q.44) Ans: B

Exp:

kaziranga national park

- Kaziranga National Park lies partly in Golaghat District and partly in Nagaon District of Assam. It is the oldest park in Assam covering an area of 430 sq km along the river Brahmaputra in the North and the Karbi Anglong hills in the South. National Highway 37 passes through the park area. It is a world heritage site that is famous for the Great Indian one-horned rhinoceros. It was declared a National Park in 1974. It is recognized as an Important Bird Area by BirdLife International for the conservation of avifaunal species.
- **Biodiversity** → The landscape of Kaziranga is of sheer forest, tall elephant grass, rugged reeds, marshes & shallow pools. It is inhabited by the world's largest population of one-horned rhinoceroses, as well as many mammals, including tigers, elephants, panthers and bears, and thousands of birds.
- **Benefits** → The satellite phones will give an edge to the forest personnel over the poachers and also during emergencies like floods.
- **Services by BSNL** → The service will be provided by Bharat Sanchar Nigam Ltd. Note: The public is barred from using satellite phones in India.

Q.45) Ans: A

Exp:

- **Statement 1 is correct and statement 2 is incorrect-** A national park is an area which is strictly reserved for the betterment of the wildlife and where **human activities like forestry, grazing or cultivation are not permitted. It is associated with the habitat of wild animal species like Rhino, Tiger etc.** Some examples of national parks are Jim Corbett, Tadoba national park etc.
- An area, whether within a sanctuary or not, can be notified by the **state government to be**

constituted as a national park, by reason of its ecological, faunal, floral, geomorphological or zoological association or importance needed to for the purpose of protecting and propagating or developing wildlife there in or its environment. No human activity is permitted inside the national park except for the ones permitted by the chief wildlife warden of the state.

- **A wildlife sanctuary** is a protected area that is reserved for the conservation only of wildlife animals and plant species. Human activities like harvesting or timber, collection of minor forest products and private ownership rights are allowed. Some examples are Periyar (Kerala), Ranipur (Uttar Pradesh) etc.
- **A biosphere reserve** is a special area of land or coastal environment in which **multiple use of land is permitted** by dividing it into certain zones. The natural or core zones. The natural or core zone consists of an undisturbed and legally protected ecosystem. The buffer zone surrounds the core area and is managed to accommodate a greater variety of resources and uses strategies.
- It takes into consideration the entire ecosystem. The purpose of the formation of the biosphere reserve is to conserve in situ all forms of life, along with its support system, in its totality, so that it could serve as a referral system for monitoring and evaluating changes in natural ecosystems.

Q.46) Ans: B

Exp:

- A fraction of the leaked fuels sink to the bottom of the sea and settle on the floor. Stimulating the benthic microbes that live at this depth of the ocean is important for effective decomposition of the pollutants present there.
- An oil spill refers to any uncontrolled release of crude oil, gasoline, fuels, or other oil by-products into the environment. Oil spills can pollute land, air, or water, though it is mostly used for oceanic oil spills.
- **Major Causes** → Oil spills have become a major environmental problem, chiefly as a result of intensified petroleum exploration and production on continental shelves and the transport of large amounts of oils in vessels. Oil spills that happen in rivers, bays and the ocean most often are caused by accidents involving tankers, barges, pipelines, refineries, drilling rigs and storage facilities, but

also occur from recreational boats and natural disasters.

- Other examples: Below are some examples of other such oil-cleaning bacteria:
 - **NY3** - The "NY3" has an "extraordinary capacity" to produce rhamnolipids that can help break down oil and then degrade some of its most serious toxic compounds.
 - **Geobacter bacteria** - They were the first organisms found to oxidise organic compounds to carbon dioxide. They can convert the organic compounds in oil spills into carbon dioxide.
 - **Alcanivorax borkumensis** - This rod-shaped microbe lives in all the world's oceans with a special preference for oil-polluted areas, as it uses hydrocarbon molecules for food.
 - **Oilzapper** - Developed by TERI & partly supported by the DBT (Department of Biotechnology), the Oilzapper is essentially a cocktail of five different bacterial strains that are immobilized and mixed with a carrier material (powdered corn cob).

Q.47) Ans: C

Exp:

- **Statement 1 is incorrect : Hydrofluorocarbons** HFCs are entirely man-made organic compounds that contain fluorine and Hydrogen atoms, and are the most common type of organofluorine compound. They do not harm the ozone layer as much as the compounds they replace but they do contribute to global warming with tri fluoromethane having 11700 times the warming potential of carbon dioxide. Their atmospheric concentration and contribution to anthropogenic greenhouse gas emissions are rapidly increasing, causing international concern about their radiative forcing
- **Statement 2 is correct** : They are primarily produced for use in refrigeration, air-conditioning, insulating foams and aerosol propellants, with minor uses as solvents and for fire protection. HFC, climate damaging refrigerants are used in air-conditioning refrigeration foams and aerosols as replacement for many ozone depleting substances . ODS are being phased out under Montreal protocol of 1987. Phasing out of ODS is important to protect the stratospheric ozone layer.

- **Statement 3 is correct** : HFCs were developed to replace stratospheric ozone-depleting substances (ODS) that are currently being phased out under the Montreal Protocol on Substances that Deplete the Ozone Layer. Though HFCs do not impact the ozone layer, many HFCs are very powerful greenhouse gases and a substantial number are short-lived climate pollutants with a lifetime of between 15 and 29 years in the atmosphere.

Reference-

<https://pib.gov.in/PressReleasePage.aspx?PRID=1746946>

Q.48) Ans: C

Exp:

- **Statement 1 is incorrect-** 'Arunachal's hornbill is inspiring — from forest guards to tribes, people protect nature'. The great hornbill (*Buceros bicornis*) is also known as the concave-casqued hornbill, great Indian hornbill or great pied hornbill. It is one of the larger members of the hornbill family. It is found in the Indian subcontinent and Southeast Asia. The most prominent feature of the Great Hornbill is the bright yellow and black casque (hard structure on the head of some birds) on top of its massive bill. Festival - The Hornbill Festival celebrated in Nagaland is named after the bird Hornbill, which is the most revered and admired bird for the Nagas. Cultural Symbol - They are the cultural symbols of some ethnic communities in the northeast, specifically the Nyishi of Arunachal Pradesh. State Bird - Great Hornbill is the state bird of Arunachal Pradesh and Kerala. Conservation Status - IUCN Status: Vulnerable, Indian Wildlife Protection Act, 1972: Schedule I.
- **Statement 2 is correct-** Dragonfly species (Anisoptera) are characterized by long bodies with two narrow pairs of intricately veined, membranous wings that, while generally transparent, may have coloured markings. Unlike damselflies, the front and rear wing pairs are shaped differently. In addition, dragonflies rest with their wings spread horizontally, rather than held vertically against each other. Dragonflies have a more powerful build and are generally much stronger fliers than damselflies. Dragonflies indicated current land use while damselflies indicated time-lagged land use.
- **Statement 3 is correct-** Large species allowed evaluating recent reduction of original vegetation.

Small species provided information of historic habitat modification. Individual species did not reflect ecological integrity. Some species' combinations indicate environmental degradation. NTCA in collaboration with the State Forest Departments, Conservation NGOs and coordinated by the Wildlife Institute of India (WII), has conducted a National assessment for the "Status of Tigers, Co-predators, Prey and their Habitat" every four years since then. The first status assessment of 2006 was peer reviewed by international carnivore experts and the IUCN. The methodology (vide Methodology Chapter of the current report for more detail) used for these 2 assessments was standardized after a pilot survey conducted in about 20,000 km area of Satpura-Maikal landscape of Central India.

Reference- Onlyias

Q.49) Ans: A

Exp:

- **Statement 1 is correct-** Organisms occupy a place in the natural surroundings or in a community according to their feeding relationship with other organisms.
- **Statement 2 is incorrect-** based on the source of their nutrition or food, organisms occupy a specific place in the food chain that is known as their trophic level. Producers belong to the first trophic level, herbivores (primary producers) to the second and carnivores (secondary consumer) to the third.
- The important point to note is that the amount of energy decreases at successive trophic levels. When an organism dies, it is converted to detritus or dead biomass that serves as an energy source for decomposers. Organisms at each trophic level depend on those at the lower trophic level for their energy demand.

Q.50) Ans: D

Exp:

- **International Energy Agency-** Established in 1974 as per the framework of the OECD, IEA is an autonomous intergovernmental organisation. Its mission is guided by four main areas of focus: energy security, economic development, environmental awareness and engagement worldwide. Headquarters: Paris, France.

- **Roles and functions:** Established in the wake of the 1973-1974 oil crisis, to help its members respond to major oil supply disruptions, a role it continues to fulfil today. IEA's mandate has expanded over time to include tracking and analyzing global key energy trends, promoting sound energy policy, and fostering multinational energy technology cooperation.
- **Composition and eligibility:** It has 30 members at present. The IEA family also includes eight association countries. A candidate country must be a member country of the OECD. But all OECD members are not IEA members.
- **Reports:**
 - Global Energy & CO2 Status Report.
 - World Energy Outlook.
 - World Energy Statistics.
 - World Energy Balances.
 - Energy Technology Perspectives
- **World energy outlook-** Published every year based on objective data and dispassionate analysis, The World Energy Outlook (WEO) provides critical analysis and insights on trends in energy demand and supply, and what they mean for energy security, environmental protection and economic development.
- Reference: <https://www.iea.org/topics/world-energy-outlook>

Q.51) Ans: A

Exp:

- **Statement 1 is correct-** Black carbon (BC) is a short-lived climate pollutant. Industry [primarily brick kilns] and residential burning of solid fuel together account for 45–66% of Black carbon emissions, On-Road diesel fuels (7–18%) and Open burning (less than 3% in all seasons).
- **Statement 2 is incorrect-** It is the second-largest contributor to warming the planet after carbon dioxide (CO2).
- **Statement 3 is correct-** It absorbs solar energy and warms the atmosphere. When it falls to earth with precipitation, it darkens the surface of snow and ice. Black carbon (BC) deposits produced by human activity accelerate the pace of glacier and snow melt in the Himalayan region.
- **Statement 4 is incorrect-** Thus reducing their albedo (the reflecting power of a surface), and warming the snow, resulting in the faster glacial melting. However, unlike other greenhouse gas emissions (GHGs), BC is quickly washed out and can

be eliminated from the atmosphere if emissions stop. Deposits of Black Carbon act in two ways hastening the pace of glacier melt, by decreasing surface reflectance of sunlight and by raising air temperature.

• **Other types of Carbon:**

- Blue Carbon - It refers to coastal, aquatic and marine carbon sinks held by vegetation, marine organisms and sediments.
- Green Carbon - It is the carbon that is stored in terrestrial ecosystems such as forests, pastures and soils.
- Brown Carbon - It is a light-absorbing particle in the Earth's atmosphere that has the unique characteristics of both cooling the planet's surface and warming its atmosphere.

Q.52) Ans: A

Exp:

- **Statement 1 is correct : Saltwater crocodiles-** Scientific Name - *Crocodylus porosus*. It is considered as the Earth's largest living crocodile species. The estuarine crocodile is infamous globally as a known man-eater. Habitat - It is found in Odisha's Bhitarkanika National Park, the Sundarbans in West Bengal and the Andamans and Nicobar Islands. It is also found across Southeast Asia and northern Australia.
- **Protection Status - IUCN Red List: Least Concern, Wild Life (Protection) Act, 1972: Schedule I, CITES: Appendix I** (except the populations of Australia, Indonesia and Papua New Guinea, which are included in Appendix II). **Threats** - Illegal hunting, habitat loss, and antipathy toward the species because of its reputation as a man-eater continues to put pressure on the population. In 1975, the Ministry of Environment, Forest and Climate Change and the United Nations Development Programme (UNDP) started a crocodile breeding and rearing project. Efforts have increased the number of crocodiles in BNP to 1,768 in January 2021 from just 96 in 1974.
- **Statement 2 is incorrect : mahseer-** Mahseer, one of the most iconic freshwater fishes of Asia is being imperilled by misidentification, release into the wrong waters, dam construction etc. Mahseer is a term used to refer to several species of freshwater fishes in Asia. Mahseers are fishes of the genus *Tor*. It is prized as a game fish and also as a food fish. Overfishing, habitat loss, pollution, dam

construction etc. are leading to the decline in mahseer population.

- **Statement 3 is correct : Gharial-** It is a species of Asian crocodilian distinguished by their long, thin snouts which resembles a pot. They prefer to live in riverine habitats with deep, clear, fast-moving water and steep, sandy banks. It was once found across Bangladesh, Bhutan, India, Myanmar, Nepal and Pakistan. However, currently it survives in several severely fragmented populations in India and Nepal. In India, Gharials are present in Son River, Girwa River, the Ganges, Mahanadi River and the Chambal River. **Protection Status - IUCN Red list: Critically Endangered, Wild Life (Protection) Act, 1972: Schedule I, CITES: Appendix I.** Threat- Construction of Dam, barrages, and water abstraction, entanglement in fishing nets, River bed cultivation and sand mining. Breeding Centres of Kukrail Gharial Rehabilitation Centre in Lucknow, Uttar Pradesh, National Chambal Sanctuary (Gharial Eco Park, Madhya Pradesh).
- **Statement 4 is incorrect-** A tree with an estimated age of **50 years** or more shall be defined as a heritage tree. It may belong to specific species which will be notified from time to time. Method to determine the age of the tree: The environment department in consultation with the forest department will issue guidelines to determine the age of the tree.

Q.53) Ans: C

Exp:

- **Statement 1 is correct-** Global assessment report (GAR) is released by the oUN Office for Disaster Risk Reduction (UNDRR).
- **Statement 2 is correct-** The report explores the systemic nature of drought and its impacts on the achievement of the Sendai Framework for Disaster Risk Reduction, the SDGs, and human and ecosystem health and wellbeing. Globally Around 700 million people are at risk of being displaced as a result of drought by 2030. Two-third of the world will be under water-stressed conditions by 2025. In India The effect of severe droughts on India's gross domestic product (GDP) is estimated at 2 The Deccan region sees the highest frequency (>6%) of severe droughts in all of India.
- **The United Nations Office for Disaster Risk Reduction (UNISDR):** The UNISDR was established in 1999 as a dedicated secretariat to facilitate the implementation of the International Strategy for

Disaster Reduction (ISDR). Headquartered at Geneva, Switzerland. It is mandated to serve as the focal point in the United Nations system for the coordination of disaster reduction and to ensure synergies among the disaster reduction activities. It is an organisational unit of the UN Secretariat and is led by the UN Special Representative of the Secretary General for Disaster Risk Reduction (SRSG).

Q.54) Ans: B

Exp:

- **About pygmy hog-** Scientific Name : 'Porcula Salvania'. It is one of the very few mammals that build its own home, or nest, complete with a 'roof'. It is also an indicator species. Its presence reflects the health of its primary habitat, the tall, wet grasslands. of the region.
- **Habitat** - It thrives in wet grassland. Once found along a narrow strip of tall and wet grassland plains on the Himalayan foothills – from Uttar Pradesh to Assam, through Nepal's terai areas and Bengal's duars. Today found mainly in a small population in Assam.
- **Protection Status** -
 - o IUCN Red List: Endangered
 - o CITES: Appendix I. Wildlife (Protection) Act, 1972: Schedule I. Threats - Habitat (grassland) loss and degradation, and illegal hunting
- **Pygmy Hog Conservation Programme (PHCP)**- The PHCP is a collaboration among Durrell Wildlife Conservation Trust of UK, Assam Forest Department, Wild Pig Specialist Group of International Union for Conservation of Nature (IUCN) and Ministry of Environment, Forest and Climate Change.

Extra edge by Onlyias

- **Indian peafowl-** The Indian peafowl is a native of India and some parts of Pakistan and Sri Lanka. The Arakan hills prevented their spread further east while the Himalayas and the Karakoram did so northwards. As our national bird, the peacock has the utmost level of legal protection. It comes under Section 51 (1-A) of Schedule I of the Wild (Life) (Protection) Act, 1972. The primary cause of their decline was poaching. Our laws allow the collection of shed feathers of peacocks, but once the feathers are being used on artefacts or other items, it is difficult to judge whether they were shed or pulled out.

- **Himalayan tahr-** The Himalayan tahr is a relative of the wild goat and is specially adapted to life on the rugged mountain slopes of the Himalayas, extending from the montane to alpine zones. Tahrs are predominantly grazers, feeding on grasses and herbs, but they do browse the leaves of shrubs particularly when pastures are snow-covered. The major threats in China are uncontrolled hunting and deforestation. In India, Himalayan tahr is sometimes hunted for meat, and there is apparently significant competition with livestock for summer grazing in some areas.
- **Mishmi takin-** Takin is the national animal of Bhutan and this species is native to India, Myanmar and China. Mishmi takin (*Budorcas taxicolor taxicolor*) is a species of goat-antelope found in the hills of Arunachal Pradesh and Sikkim at an altitude of 2000-3000 metres. It is listed under Schedule I of the Indian Wildlife (Protection) Act, 1972 and is also classified as endangered by the International Union for Conservation of Nature (IUCN). Yet, only scant information exists on the population, distribution, ecology and behaviour of this endangered species.

Q.55) Ans: A

Exp:

About UNCCD-

- Established in 1994. It is the sole legally binding international agreement linking environment and development to sustainable land management. It is the only convention stemming from a direct recommendation of the Rio Conference's Agenda 21. The Convention addresses specifically the arid, semi-arid and dry sub-humid areas, known as the drylands, where some of the most vulnerable ecosystems and peoples can be found. The Convention's 197 parties work together to improve the living conditions for people in drylands, to maintain and restore land and soil productivity, and to mitigate the effects of drought.
- The UNCCD is particularly committed to a bottom-up approach, encouraging the participation of local people in combating desertification and land degradation. The UNCCD secretariat facilitates cooperation between developed and developing countries, particularly around knowledge and

technology transfer for sustainable land management.

- **14th Session of COP-14 of UNCCD:** India had for the first time hosted the 14th session of the Conference of Parties(COP-14) of the UNCCD at Greater Noida in 2019. The theme of the Conference was 'Restore land, Sustain future'. During the conference, the Delhi Declaration was issued. The declaration called for better access over land and emphasised gender-sensitive transformative projects.
- **Delhi Declaration:** Commitment for a range of issues, including gender and health, ecosystem restoration, taking action on climate change, private sector engagement, Peace Forest Initiative and recovery of five million hectares of degraded land in India.
- Reference: <https://www.unccd.int/>

Q.56) Ans: A

Exp:

- **Statement 1 is correct-** The central government can take the following measures for the protection of the environment and for the prevention, control and abatement of pollution.
- **Statement 2 is correct-** Under the environmental protection act, 1986 the power of the central government includes:
 - Laying down the standards for environmental quality.
 - Laying down procedures and safeguards for the handling of hazardous materials.
 - Planning and execution of programmes to prevent, control and abate environmental pollution on an all India basis.
 - Making rules in respect of all matters pertaining to environmental protection as well as to pollution.
 - Restricting areas in which industries or other establishments are to be set up.
- **Statement 3 is incorrect-** The forest conservation act, 1981 was amended in 1988. The special feature of this act is that it compels a state government to take 'prior approval' of the government of India if it de-reserved a reserved forest, that is to say, if the state government wants to use forest land for a non-forest purpose.
- Reference: Onlyias

Q.57) Ans: B

Exp:

- **Manas National Park:** Manas national park is a UNESCO World Heritage Site, a Project Tiger reserve, an elephant reserve and a biosphere reserve. It is located in the Himalayan foothills in Assam. It is contiguous with the Royal Manas National Park in Bhutan. The name of the park originated from the Manas River.
- **Significance:** The park is known for its rare and endangered endemic wildlife such as the Assam roofed turtle, hispid hare, golden langur and pygmy hog. It is also famous for its population of wild water buffalo. Pagrang is the only forest village located in the core of the Manas national park.
- **River:** The Manas river flows through the west of the park. Manas is a major tributary of Brahmaputra river. The river is named after the serpent goddess Manasa.



Reference: Onlyias

Q.58) Ans: C

Exp:

- **Pair 1 is correct-**
Barnadi wildlife sanctuary- Bornadi Wildlife Sanctuary (BWS) is located in northern Assam's Baksa and Udalguri districts bordering Bhutan. The Sanctuary is bordered by the Barnadi river and Nalapara river to the west and east respectively. About 60% of the Barnadi wildlife sanctuary is reported to be grassland, most of it is now grassy woodland. The main Forest types are Tropical Moist Deciduous along the northern edge of the Sanctuary, and mixed scrub and grassland in the southern part with few trees. The vegetation has been drastically modified by human activity. Most of the natural vegetation has been cleared and replaced by commercial plantations of Bombax ceiba, Tectona grandis and Eucalyptus and by thatch grasses (mostly Saccharum, with some Phragmites and Themeda).



Other Protected Areas in Assam:

- Dibru-Saikhowa National Park,
- Manas National Park,
- Nameri National Park,
- Rajiv Gandhi Orang National Park.
- Kaziranga National Park.

• Pair 2 is correct-

Ramgarh vishdhari wildlife sanctuary- This will be the 52nd Tiger Reserve of India. This Sanctuary is located at a distance of 45 Km from Bundi City on Bundi-Nainwa Road near Village Ramgarh, District Bundi, Rajasthan. It was notified in the Year 1982 and is spread over an area of 252.79 Sq. Km. Its flora consists of Dhok, Khair, Salar, Khirni trees with some Mango and Ber trees. The Fauna consists of birds and animals like Leopard, Sambhar, Wild boar, Chinkara, Sloth bear, Indian Wolf, Hyena, Jackal, Fox, deer and Crocodile.

Other Protected Areas in Rajasthan:

- Ranthambore tiger reserve
- Mukundra hills tiger reserve
- Sariska tiger reserve
- Desert National Park, Jaisalmer
- Keoladeo National Park, Bharatpur
- Sajjangarh wildlife sanctuary, Udaipur
- National Chambal Sanctuary

• Pair 3 is not correct-

Periyar wildlife sanctuary- It falls in the districts of Idukki and Pathanamthitta in Kerala. Declared a Sanctuary during 1950 and declared as Tiger Reserve during 1978. It gets its name from the River Periyar which has its origin deep inside the reserve. The major rivers through the reserve are Mullayar and Periyar. The sanctuary comprises tropical evergreen, semi evergreen, moist deciduous forests and grasslands. Mammals: Tiger, Elephant, Lion-tailed macaque, Nilgiri Tahr etc. Birds: Darters, Cormorants, Kingfishers, the great Malabar Hornbill and racket-tailed Drongos. Reptiles: Monitor Lizards, Python, King Cobra etc. There are six tribal communities nestled inside the reserve such as Mannans, Paliyans, Malayarayangans, Mala Pandarams, Uralis and Ulladans.

• Pair 4 is not correct-

Dandeli wildlife sanctuary- The Dandeli Wildlife Sanctuary is located at Uttara Kannada District in Karnataka. The sanctuary is a bird watcher's paradise, housing nearly 200 species of birds, most famous for the great Indian hornbill and the Malabar pied hornbill. It is also the only known tiger reserve in India to report frequent sightings of the elusive black panther. It is also known to house the Indian sloth bear, the Indian pangolin, the giant Malabar squirrel, dhole, the Indian jackal and the muntjac (barking deer). The king cobra

and the mugger crocodile (Indian crocodile) are the prime reptilians in the sanctuary.

Q.59) Ans: B

Exp:

Nilgiri biosphere reserves

- **Nilgiri Biosphere Reserve** was included in UNESCO World network of Biosphere Reserves in the year 2000. The Nilgiri Biosphere Reserve has the distinction of being the first biosphere reserve in India. It is located in the **Western Ghats**. The Nilgiri Biosphere Reserve lies in the biogeographic region of the **Malabar rain forest**. Nilgiri Biosphere reserve core areas major parts spread over Kerala and Tamil Nadu States, and this include evergreen, semi evergreen, moist deciduous montane sholas and grassland types of vegetation. Nilgiri Biosphere reserve core area spread over the State of Karnataka and consists mostly of dry deciduous forests and a few patches of moist deciduous, semi evergreen and scrub jungles.
- Protected areas present within this reserve. **Nagarhole National Park, Bandipur National Park, Wayanad Wildlife Sanctuary, Mudumalai Tiger reserve and Wildlife Sanctuary, Mukurthi National Park, Silent Valley National Park.**
- The Nilgiri Biosphere Reserve comprises substantial unspoilt areas of natural vegetation thus contributing to highest biodiversity. The altitude and climatic gradients support the different vegetational types. In 1992 Nagarhole National Park was renamed as "Rajiv Gandhi (Nagarhole) National Park" to mark the first death anniversary of Late Prime Minister Shri. Rajiv Gandhi. Nagarhole National Park is partly situated in – Kodagu district and partly in Mysore district.

Reference-

[Mysuru://en.esco.org/biosphere/aspac/nilgiri](http://en.esco.org/biosphere/aspac/nilgiri)

Q.60) Ans: B

Exp:

- **Statement 1 is incorrect:** Mission Innovation (MI) is a global initiative of 24 countries and the European Commission (on behalf of the European Union). Mission Innovation is a global initiative catalysing a decade of action and investment in research, development and demonstration to make clean energy affordable, attractive and accessible for all. These resources have dramatically accelerated the availability of the



advanced technologies that will define a future global energy mix which is clean, affordable, and reliable.

- **Statement 2 is correct:** This will accelerate progress towards the Paris Agreement goals and pathways to net zero. Commitment by all members to seek to double their clean energy innovation investments over five years in selected priority areas. Each member according to its own priorities, policies, processes, and laws independently determines the best use of its funding and defines its own Research & Development priorities and path to reach the doubling goal.

Source : <https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1590258>

Q.61) Ans: C

Exp:

Clean Energy International Incubation Centre

- The Clean Energy International Incubation Centre (CEIIC), which is Social Alpha's Energy Lab, is a joint initiative of Tata Trusts and the Government of India.
- It has been set up for promoting innovations in the energy space and has become the first International Incubator in India under Mission Innovation.
- The Incubator is designed to offer complete "lab to market" incubation support to clean energy enterprises, both Indian and International, which can bring about deep and irreversible social and environmental impact.
- CEIIC supports the incubatees by providing last-mile connectivity and end-use deployment of successful research outputs.
- CEIIC focuses on three core areas for innovation – Energy Access, Energy Efficiency and Breakthrough Clean Technology. The convergence of these holds the potential to transform the way the world produces and consumes energy.

Source: <http://ceiic.socialalpha.org/>

Q.62) Ans: C

Exp:

- **Statement 1 is correct:** Clean Energy Research Initiative is a technology Development Scheme. The initiative aims to develop national research competence to drive down the cost of clean energy

through pre-competitive translational research, oriented research led disruptive innovations & human and institutional capacity development. It intends to nurture S & T-led breakthroughs to feed the national mission on solar energy.

- **Statement 2 is correct:** It helps to develop India centric innovations developed around user needs and forge collaboration between industry and academics as much as possible and gain value for such collaborations and also develop critical mass of researchers to meet requirements of R&D professionals for clean energy.

Source : <https://dst.gov.in/clean-energy-research-initiative>

Q.63) Ans: C

Exp:

- **Statement 1 is correct:** The United States and India launched the Partnership to Advance Clean Energy (PACE), which is working to accelerate inclusive, low carbon growth by supporting research and deployment of clean energy technologies. Both countries are committed to strengthening and expanding PACE through a series of initiatives. India in the multilateral Clean Energy Ministerial forum.
- **Statement 2 is correct:** Over the past year, India has revised its renewable energy target to 175 GW by 2022. The national solar target was scaled up by five times, reaching 100 GW by 2022, of which 40 GW is expected to come from solar rooftop. The activities of the PACE initiative are aligned to support India's ambitious clean energy targets.

Source:

<https://www.climatelinks.org/resources/partnership-advance-clean-energy-deployment-pace-d-india>

Q.64) Ans: A

Exp:

- **Statement 1 is correct:** Lignocellulosic biomass can be utilized to produce ethanol, a promising alternative energy source for the limited crude oil. There are mainly two processes involved in the conversion: hydrolysis of cellulose in the lignocellulosic biomass to produce reducing sugars, and fermentation of the sugars to ethanol.
- **Statement 2 is incorrect:** The cost of ethanol production from lignocellulosic materials is relatively high based on current technologies, and the main challenges are the low yield and high cost



of the hydrolysis process. Pretreatment of lignocellulosic materials to remove lignin and hemicellulose can significantly enhance the hydrolysis of cellulose. Optimization of the cellulase enzymes and the enzyme loading can also improve the hydrolysis.

Source:

<https://www.google.co.in/search?q=Lignocellulose+Hydrolysis+Technology:&ie=UTF-8&oe=UTF-8&hl=en-in&client=safari>

Q.65) Ans: B

Exp:

- Fly ash is a by-product of coal-based power generation. It is a fine powder with substantial amounts of oxides of silica, aluminum and calcium. It also contains traces of Arsenic, Boron, Chromium, lead etc. which leads to air and water pollution if disposed of on land. With low grades of Indian coal, its ash content is as high as 30-45% in comparison to imported coal with 10-15%.
- **Statement 1 is incorrect-** First fly ash notification was issued in 1999 to ensure 100% fly ash utilization in India by 2009. This was followed by a similar notification in 2016. The current notification aims to achieve the objective of 100% utilization in 3 to 5 years. Existing provisions allow TPPs to fully utilize fly ash in a four-year cycle in a staggered manner. The new policy will follow a three-year cycle for 100% utilization of Fly-ash with a grace period of a year if the percentage of ash utilization is between 60-80% and two years if it is below 60%.
- **Statement 2 is correct-** The non-complying power plants will provide ash free-of-cost to agencies engaged in construction activities within a 300 km radius with all transportation cost to be borne by TPPs.
- **Role of Central Pollution Control Board-** A committee under the chairmanship of CPCB chairman will examine, review and recommend eco-friendly ways on fly ash utilization. Also, CPCB will have real-time data on ash availability.
- Reference:
<https://pib.gov.in/Pressreleaseshare.aspx?PRID=1540124>

Q.66) Ans: D

Exp:

- **Statement 1 is correct:** The purpose of the bill is to regulate the use of DNA information for establishing the identity of people. These profiles are then meant to guide law enforcement agencies in investigations. The committee has underlined that it is important that state-of-the-art technologies are used in the criminal justice system, but this must be done without infringing constitutional rights, especially the Right to Privacy. Although DNA technology can help law enforcement agencies, in solving crimes, the government must assuage apprehensions over the use of the DNA Technology Bill, 2019.
- **Statement 2 is correct:** The Bill mandates the permanent storage of DNA data obtained under all categories, including those obtained from a criminal investigation. This implies that an individual's data, if picked up from a crime scene, may find its way to the database without their explicit knowledge. It is possible for suspects who have been acquitted to put in a written request for removing their samples from the data bank.
- **Statement 3 is correct:** It also has the provisions to establish National and Regional DNA Data Banks for the maintenance of the national forensic DNA database for the purposes of identification of the aforementioned category of persons. This will aid in scientific up-gradation and streamlining of the DNA testing activities in the country. Establishment of the National and Regional DNA Data Banks, as envisaged in the Bill, will assist in forensic investigations.

Source: <https://prsindia.org/billtrack/the-dna-technology-use-and-application-regulation-bill-2019>

Q.67) Ans: B

Exp:

BBX11

- It is a newly identified gene that facilitates greening of plants by regulating levels of protochlorophyllide — an intermediate in the biosynthesis of chlorophyll. The synthesis of chlorophyll in plants is a lengthy, multi-step process.
- When a seedling emerges from under the soil it must quickly synthesise chlorophyll to start supporting its own growth. In order to facilitate quick synthesis of chlorophyll, plants make a precursor of chlorophyll called 'protochlorophyllide' in the dark, which glows red when blue light is shone on the plant. As soon as

the plant comes out into the light from under the soil, light-dependent enzymes convert protochlorophyllide to chlorophyll.

- Using genetic, molecular and biochemical techniques, researchers found a mechanism where two proteins oppositely regulate the 'BBX11' gene to maintain optimum levels of 'BBX11'. So that the amount of protochlorophyllide synthesised needed to be proportional to the number of enzymes available to convert them to chlorophyll.

Source:

<https://www.thehindu.com/news/states/iiser-scientists-identify-the-gene-that-greens-plants/article33420460.ece>

Q.68) Ans: C

Exp:

Genetically Modified Crops

- GM is a technology that involves inserting DNA into the genome of an organism. GM plants give crops some desired properties like – pest resistance, herbicide-tolerance, drought resistance etc. New DNA is transferred into plant cells. Usually, the cells are then grown in tissue culture where they develop into plants. The seeds produced by these plants will inherit the new DNA.
- Currently, the only GM crop permitted for cultivation in India is Bt Cotton. But cultivation of illegal GM crops (Brinjal, Soyabean etc) has been reported from across states.
- There is a well established regulatory framework for approval of GM Crops as per "Rules for the Manufacture/Use/Import/Export and Storage of Hazardous Microorganisms, Genetically Engineered Organisms or Cells, 1989" under the Environment (Protection) Act, 1986.
- There are certain concerns regarding GM crops for Human health (like allergic reaction, gene transfer etc.), Environment (like introduction of engineered genes into wild populations, loss of biodiversity. GEAC is the apex body that allows for commercial release of GM crops.

Source: <https://royalsociety.org/topics-policy/projects/gm-plants/what-is-gm-and-how-is-it-done/>

Q.69) Ans: B

Exp:

- Statement 1 is incorrect:** The Genetic Engineering Appraisal Committee (GEAC) functions in the Ministry of Environment, Forest and Climate Change (MoEF&CC). GEAC is chaired by the Special Secretary/Additional Secretary of MoEF&CC and co-chaired by a representative from the Department of Biotechnology (DBT). Presently, it has 24 members and meets every month to review the applications in the areas indicated above.
- Statement 2 is correct:** As per Rules, 1989, it is responsible for appraisal of activities involving large scale use of hazardous microorganisms and recombinants in research and industrial production from the environmental angle. The committee is also responsible for appraisal of proposals relating to release of genetically engineered (GE) organisms and products into the environment including experimental field trials.

Source : <http://geacindia.gov.in/about-geac-india.aspx>

Q.70) Ans: C

Exp:

- Statement 1 is correct:** Nanobiotechnology is a discipline in which tools from nanotechnology are developed and applied to study biological phenomena. Nanoscience is the study of materials which are in nanoscale range. Conversion of any material in nanoscale results in alteration of its physicochemical, biological, mechanical, optical, electronic, etc. properties. These newly acquired (novel) properties of the materials due to conversion into a nanoscale can be utilized for different useful activities.
- Statement 2 is correct:** Nanobiotechnology enables technology relevant for diverse sectors, such as chemicals, consumer products, health, energy, various other industries and the environment. The use of this technology is increasing exponentially in the pharmaceutical sector. Department of biotechnology has initiated funding R&D in this area since 2007. In the initial year's exploratory type of projects were funded. The approach at the present moment is to found application/translation oriented projects. Key objective of the programme is to advance research and promote innovation through applications of nanobiotechnology to address issues in health, energy, agriculture and environment.



Source: <https://dbtindia.gov.in/schemes-programmes/research-development/nanobiotechnology>

Q.71) Ans: C

Exp:

Genome editing Technologies

- Genome editing (also called gene editing) is a group of technologies that give scientists the ability to change an organism's DNA. These technologies allow genetic material to be added, removed, or altered at particular locations in the genome.
- Several approaches to genome editing have been developed. A recent one is known as CRISPR-Cas9, which is short for clustered regularly interspaced short palindromic repeats and CRISPR-associated protein 9.
- The CRISPR-Cas9 system has generated a lot of excitement in the scientific community because it is faster, cheaper, more accurate, and more efficient than other existing genome editing methods.
- CRISPR-Cas9 was adapted from a naturally occurring genome editing system in bacteria. The bacteria capture snippets of DNA from invading viruses and use them to create DNA segments known as CRISPR arrays.
- The CRISPR arrays allow the bacteria to "remember" the viruses (or closely related ones). If the viruses attack again, the bacteria produce RNA segments from the CRISPR arrays to target the viruses' DNA. The bacteria then use Cas9 or a similar enzyme to cut the DNA apart, which disables the virus.

Source:

<https://medlineplus.gov/genetics/understanding/genomicresearch/genomeediting/>

Q.72) Ans: A

Exp:

- **Statement 1 is correct:** Stem cells are the body's raw materials — cells from which all other cells with specialized functions are generated. Under the right conditions in the body or a laboratory, stem cells divide to form more cells called daughter cells. These daughter cells either become new stem cells (self-renewal) or become specialized cells (differentiation) with a more specific function, such as blood cells, brain cells, heart muscle cells

or bone cells. No other cell in the body has the natural ability to generate new cell types.

- **Statement 2 is incorrect:**

These stem cells come from

- **Embryonic stem cells.** These stem cells come from embryos that are three to five days old. At this stage, an embryo is called a blastocyst and has about 150 cells. These are pluripotent (plo-o-RIP-uh-tunt) stem cells, meaning they can divide into more stem cells or can become any type of cell in the body. This versatility allows embryonic stem cells to be used to regenerate or repair diseased tissue and organs.
- **Adult stem cells.** These stem cells are found in small numbers in most adult tissues, such as bone marrow or fat. Compared with embryonic stem cells, adult stem cells have a more limited ability to give rise to various cells of the body. Recently Adult stem cells are currently being tested in people with neurological or heart disease.

Source:

<https://www.mayoclinic.org/tests-procedures/bone-marrow-transplant/in-depth/stem-cells/art-20048117>

Q.73) Ans: A

Exp:

Biotech-KISAN

- Biotech-Krishi Innovation Science Application Network (Biotech-KISAN) is a Department of Biotechnology, **Ministry of Science and Technology** initiative that empowers farmers, especially women farmers.
- It aims to understand the problems of water, soil, seed and market faced by the farmers and provide simple solutions to them.
- The Scheme is for farmers, developed by and with farmers, it empowers women, impacts locally, connects globally, is Pan-India, has a hub-and-spoke model and stimulates entrepreneurship and innovation in farmers.

Objectives and Scope of coverage

- Biotech-Krishi Innovation Science Application Network (Biotech-KISAN) is being implemented in 15 agro-climatic zones of India in phased manner with the following objectives:
 - Linking available science and technology to the farm by first understanding the problem of the local farmer and providing solutions to those problems.



- The working together, in close conjunction, of scientists and farmers is the only way to improve the working conditions of small and marginal farmers.
- This programme aims to work with small and marginal farmers especially the woman farmer for better agricultural productivity through scientific intervention and evolving best farming practices in the Indian context.

Source: <https://vikaspedia.in/agriculture/policies-and-schemes/crops-related/biotech-kisan-scheme>

Q.74) Ans: C

Exp:

- **Statement 1 is correct:** The Human Microbiome is the collection of all the microorganisms living in association with the human body. These communities include eukaryotes, archaea, bacteria and viruses. Bacteria in an average human body number ten times more than human cells, for a total of about 1000 more genes than are present in the human genome. Because of their small size, however, microorganisms make up only about 1 to 3 percent of our body mass (that's 2 to 6 pounds of bacteria in a 200-pound adult).
- **Statement 2 is correct:** These microbes are generally not harmful to us, in fact they are essential for maintaining health. For example, they produce certain vitamins that we do not have the genes to make, break down our food to extract nutrients we need to survive, teach our immune systems how to recognize dangerous invaders and even produce helpful anti-inflammatory compounds that fight off other disease-causing microbes. An ever-growing number of studies have demonstrated that changes in the composition of our microbiomes correlate with numerous disease states, raising the possibility that manipulation of these communities could be used to treat disease.

Source : <https://www.hmpdacc.org/overview/>

Q.75) Ans: D

Exp:

About Patent

- A patent is a form of preservation of intellectual property. It is an exclusive right granted for an invention, which is a product or a process that provides, in general, a new way of doing

something, or offers a new technical solution to a problem.

Recent Patents (Amendment) Rules, 2021

- The fastest granted patent is the one which was granted in 41 days after filing of such request. This facility of Expedited Examination system was initially provided for patent applications filed by Startups. It has been now extended to 8 more categories of Patent Applicants w.e.f. 17-09-2019 by making necessary amendments in the Patents Rules.
- These new categories include SME, Female applicants, Government Departments, Institutions established by a Central, Provincial or State Act, which is owned or controlled by the Government, Government Company, an Institution wholly or substantially financed by the Government and applicants under Patents Prosecution Highway.
- The Patent Prosecution Highway (PPH) is a set of initiatives for providing accelerated patent prosecution procedures by sharing information between some patent offices. With regard to Initiatives for Startup India program, 80% fee concession has been provided for Patent applications filed by Startups.

<https://pib.gov.in/PressReleasePage.aspx?PRID=1757202>

Q.76) Ans: B

Exp:

International Intellectual Property Index

- India ranked 40 among 53 global economies on the latest annual edition of the International Intellectual Property (IP) Index.
- Released annually by the US Chamber of Commerce Global Innovation Policy Centre (GIPC), the Index evaluates Intellectual Property rights in 53 global economies from patent and copyright policies to commercialisation of IP assets and ratification of international treaties.
- The overall global IP environment improved in 2020, with positive scores increasing in 32 of the 53 economies measured by the IP Index.
- This is the ninth IP index released by the GIPC. India ranked 40th in 2020, scoring 38.4 out of 100 on a set of 50 intellectual property-related indicators.
 - India's overall score has increased from 36.04 per cent (16.22 out of 45) in the seventh edition to 38.46 per cent (19.23 out of 50) in the eighth edition.



- India made a string of positive efforts which resulted in a score increase because of stronger enforcement efforts and precedent-setting court cases involving copyright and trademark infringement.

Source: https://www.business-standard.com/article/current-affairs/india-ranks-40th-on-international-intellectual-property-index-121032301401_1.html

Q.77) Ans: B

Exp:

- **Statement 1 is correct:** A patent is granted for an invention which is a new product or process that meets conditions of novelty, non-obviousness and industrial use. Patents in India are governed by "The patent Act 1970" which was amended in 2005 to make it compliant with TRIPS.
- **Statement 2 is incorrect:** A trademark means a mark capable of being represented graphically and which is capable of distinguishing the goods or services of one undertaking from those of other undertakings. Trademarks in India are governed by Trade Marks Act 1999 which was amended in 2010. Trade Mark Rules, 2017 has been notified which provides for ease of filing trademarks, rationalised trademark fee etc. Modalities for determining well-known trademarks has been introduced for the first time
- **Statement 3 is incorrect:** An industrial design consists of the creation of a shape, configuration or composition of pattern or color, or combination of pattern and color in three dimensional form containing aesthetic value. Designs in India are governed by "The Designs Act 2000".

Source:

<https://www.investopedia.com/articles/investing/111014/patents-trademarks-and-copyrights-basics.asp>

Q.78) Ans: C

Exp:

- **Statement 1 is correct:** Patent pools can be defined as an agreement between two or more patent owners to license one or more of their patents to one another or to third parties. Often, patent pools are associated with complex technologies that require complementary patents in order to provide efficient technical solutions.

Generally, these patent pools cover mature technologies.

- **Statement 2 is correct:** On the one hand, patent pools may have positive effects on competition and innovation. By sharing intellectual property assets, companies may develop new products and reduce their transaction costs. On the other hand, under specific circumstances, patent pools may provide an opportunity for a possible anti-competitive behavior. In addition, there may be competition-related concerns regarding the licensing practices and restrictions they entail. The so-called 'patent thickets' can lead to increased transaction costs and to chilling effects on the development of new products.

Source

<https://www.thehindubusinessline.com/opinion/patent-pooling-a-covid-success-story/article31891866.ece>

Q.79) Ans: C

Exp:

- **Statement 1 is correct:** The Patent Law Treaty (PLT), adopted by WIPO. The PLT is the product of several years of multilateral negotiations on harmonizing global patent systems. The PLT harmonizes certain patent application procedures in order to reduce or eliminate formalities and the potential for loss of rights. The Patent Law Treaty does not harmonize substantive patent law, that is, the laws of each country that set forth the conditions that must be met in order to receive a patent for an invention in that country.
- **Statement 2 is correct:** The The Patent Law Treaty:
 - simplifies and minimizes patent application requirements to obtain a filing date;
 - imposes a limit on the formal requirements that Contracting Parties may impose;
 - eases representation requirements for formal matters;
 - provides a basis for the electronic filing of applications;
 - provides relief with respect to time limits that may be imposed by the Office of a Contracting Party and reinstatement of rights where an applicant or owner has failed to comply with a time limit and that failure has the direct consequence of causing a loss of rights; and



- provides for correction or addition of priority claims and restoration of priority rights.

As of May 2021, the PLT had 43 contracting states, while 59 states and the European Patent Organisation have signed the treaty. India is not a signatory to the treaties.

Source : <https://www.uspto.gov/ip-policy/patent-policy/patent-law-treaty>

Q.80) Ans: D

Exp:

IPrism

- The Cell for IPR Promotion and Management (CIPAM), Department for Promotion of Industry and Internal Trade, in collaboration with ASSOCHAM and ERICSSON India, has launched the second edition of 'IPrism'.
- It is an Intellectual Property (IP) competition for students of schools, polytechnic institutes, colleges and universities. Aiming to foster a culture of innovation and creativity in the younger generation, the competition will provide young creators an opportunity to see their creations recognized on a national platform. This competition is for residents of India only.

About CIPAM

- It has been created under the aegis of Department for Promotion of Industry and Internal Trade to take forward the implementation of the National IPR Policy 2016.
- It assists in simplifying and streamlining of IP processes, apart from undertaking steps for furthering IPR awareness, commercialization and enforcement.

source:

<https://pib.gov.in/Pressreleaseshare.aspx?PRID=1566240>

Q.81) Ans: D

Exp:

- The Paris Convention, adopted in 1883, applies to industrial property in the widest sense, including patents, trademarks, industrial designs, utility models, service marks, trade names, geographical indications and the repression of unfair competition.
- This international agreement was the first major step taken to help creators ensure that their

intellectual works were protected in other countries.

Source: <https://www.wipo.int/treaties/en/ip/paris/>

Q.82) Ans: C

Exp:

National IPR Policy, 2016

- The National Intellectual Property Rights (IPR) Policy 2016 as a vision document to guide future development of IPRs in the country. It's clarion call is "Creative India; Innovative India".
- It encompasses and brings to a single platform all IPRs, taking into account all inter-linkages and thus aims to create and exploit synergies between all forms of intellectual property (IP), concerned statutes and agencies.
- The Policy aims to push IPRs as a marketable financial asset, promote innovation & entrepreneurship, while protecting public interest. The plan will be reviewed every five years in consultation with stakeholders.
- The policy is entirely compliant with the WTO's agreement on TRIPS. Department of Industrial Policy & promotion (DIPP) is the nodal agency for all IPR issues.
- The policy retains the provisions on Compulsory Licensing (CL) as well as preventing ever-greening of drug patents (Section 3(d) of India's Patents Act). Under Indian Patents Act, a CL can be issued for a drug if the medicine is deemed unaffordable, among other conditions, & the government grants permission to qualified generic drug makers to manufacture it.

Source: <https://dpiit.gov.in/policies-rules-and-acts/policies/national-ipr-policy>

Q.83) Ans: D

Exp:

EXPORT - PROHIBITED GOODS

- The prohibited items are not permitted to be exported. An export licence will not be given in the normal course for goods in the prohibited category.
- **The following are the items prohibited in the flora and fauna category:**
- All wild animals, animal articles including their products and derivatives (excluding those for which ownership certificates have been granted and also those required for transactions for education,

scientific research and management under Wild Life (Protection) Act, 1972 including their parts and products).

- **Beef of cows, oxen and calves.**
- Beef in the form of offal of cows, oxen and calf
- Meat of buffalo (both male and female) fresh and chilled and frozen
- **Peacock Tail Feathers & its Handicrafts** and articles
- Shavings & Manufactured Articles of shavings of Shed Antlers of Chital and Sambar
- **Sea shells**
- **Wood and wood products**
- Fuel wood
- **Wood charcoal**
- Sandalwood in any form, (but excluding finished handicraft products of sandalwood, machine finished sandalwood products, sandalwood oil)
- Red Sanders wood, Value added products of Red Sanders
- Mechanical, chemical and semi chemical wood pulp

Source : Environment Book by Shankar IAS, CHAPTER - 15 PROTECTED AREA NETWORK

Q.84) Ans: D

Exp:

Sustainable Sugarcane Initiative (SSI)

- Sustainable Sugarcane Initiative (SSI) is an **innovative set of agronomic practices that involves using less seeds, raising seeds in a nursery, and following new planting methods**, with wider seed spacing, and better water and nutrient management to increase the cane yields significantly.
- SSI methods can **increase sugarcane yields by at least 20% with 30% less water** and a 25% reduction in chemical inputs.
- The SSI method of sugarcane cultivation evolved from the **principles of 'More with Less'** followed in SRI (System of Rice Intensification) and was introduced in India by the WWF-ICRISAT collaborative project in 2009.

Source : Environment book, Shankar IAS CHAPTER - 24 AGRICULTURE

Q.85) Ans: A

Exp:

About Messenger RNA (mRNA) vaccines

- **Messenger RNA (mRNA) vaccines** teach our cells how to make a protein that will **trigger an immune response inside our bodies**. It triggers an immune response inside our bodies. That immune response, which **produces antibodies, is what protects us from getting infected if the real virus enters our bodies**. The **Pfizer-BioNTech** and **Moderna** COVID-19 vaccines are **messenger RNA vaccines** – also called mRNA vaccines.
- The mRNA will enter the muscle cells and **instruct the cells' machinery to produce a harmless piece of what is called the spike protein**. The spike protein is found on the surface of the virus. After the protein piece is made, *our cells break down the mRNA and remove it*.
- Our immune system recognizes that the protein doesn't belong there. This **triggers our immune system to produce antibodies and activate other immune cells to fight off what it thinks is an infection**.
- Ultimately, our bodies have learned how to protect against future infection from the virus.
- **mRNA vaccines are synthetically made and do not need cell cultures, bacteria, or other hosts for growing it.**

Extra Edge by Only IAS

- **mRNA never enters the nucleus of the cell** where our DNA (genetic material) is located, so it cannot change or influence our genes.
- **mRNA vaccine technology** may allow for one vaccine to provide protection against multiple diseases, thus decreasing the number of shots needed for protection against common vaccine-preventable diseases.
- Beyond vaccines, **cancer research has used mRNA to trigger the immune system to target specific cancer cells**.

Other types of vaccines :

Inactivated vaccine :

- Where Disease-carrying virus or bacteria **inactivate or kill it using chemicals, heat or radiation**.
- This is the way the **flu and polio vaccines are made** – and vaccines can be manufactured on a reasonable scale.. It requires special laboratory facilities to grow the virus or bacterium safely, and can have a relatively long production time.

Live-attenuated vaccine :

It uses a living but weakened version of the virus. The **measles, mumps and rubella (MMR) vaccine and the chickenpox and shingles vaccine**, and **yellow fever** are examples of this type of vaccine. This can be manufactured at scale.

- These vaccines like this may **not be suitable for people with compromised immune systems**.

Viral vector vaccine:

- This type of vaccine uses a safe virus to deliver specific sub-parts – called **proteins** – of the **germ of interest so that it can trigger an immune response without causing disease**.
- To give instructions for making particular parts of the pathogen of interest be inserted into a safe virus. The safe virus then serves as a platform or vector to deliver the protein into the body.
- The **Ebola vaccine is a viral vector vaccine**.

Source - <https://www.thehindu.com/sci-tech/health/mrna-vaccine-manufacturers-loath-to-share-know-how/article36670261.ece>

Q.86) Ans: C

Exp:

Convalescent plasma therapy

- Recently, **Convalescent plasma** has been used worldwide to treat patients hospitalized with coronavirus disease 2019 (COVID-19) and prevent disease progression.
- **convalescent plasma represents a form of passive antibody therapy** that relies on the **transfer of pathogen-specific antibodies from a recovered patient for the purpose of preventing or treating disease**.
- Unlike vaccines and monoclonal antibodies, **convalescent plasma therapy requires limited development or infrastructure**, requiring no more than the availability of disease survivors willing to donate plasma and standard blood collection infrastructure to collect and distribute convalescent plasma.
- Convalescent **plasma can be readily used in low resource settings across the globe**. Convalescent plasma is also adaptable to changing conditions.
- It **reduces the mortality and promotes the recovery** of viral-infected patients, but in

In a few cases, patients deteriorated due to usage of an inappropriate dosage of convalescent plasma or using the therapy at an improper stage of infection.

- convalescent plasma contains different **neutralizing antibodies**, not each one of them is useful in treating the virus.

Source :

<https://m.economictimes.com/magazines/panache/contrary-to-popular-belief-blood-plasma-therapy-doesnt-help-seriously-ill-covid-patients/articleshow/86165475.cms>

Q.87) Ans: C

Exp:

About Ebola

- **Fruit bats** of the Pteropodidae family are **natural Ebola virus hosts**. Ebola **haemorrhagic fever is transmitted to people from wild animals** and spreads in the human population through human to human transmission.
- **Ebola virus was first discovered in 1976 near the Ebola River** in what is now the Democratic Republic of Congo. The **virus family Filoviridae includes three genera**: Cuevavirus, Marburgvirus, and Ebolavirus.
- Ebola is introduced into the human population through **close contact with the blood, secretions, organs or other bodily fluids of infected animals such as fruit bats, chimpanzees, gorillas, monkeys, forest antelope**.
- **Burial ceremonies that involve direct contact with the body of the deceased can also contribute to the transmission of Ebola**.
- People remain infectious as long as their blood contains the virus.
- **Early supportive care with rehydration, symptomatic treatment improves survival**. Two **monoclonal antibodies (Inmazeb and Ebanga)** were approved for the treatment of Zaire ebolavirus (Ebolavirus) infection in adults and children by the US Food and Drug Administration in late 2020.
- Ebola virus can be **transmitted From Pregnant women to babies by breast milk, and other pregnancy fluids**.

Extra Edge by Only IAS

Ebola virus

- **It's incubation period is 2 to 21 days**. A person infected with Ebola cannot spread the disease until they develop symptoms.

• **There are the following fews symptoms of EVD**

- Fever
- Fatigue
- Muscle pain
- Headache
- Sore throat
- Symptoms of impaired kidney and liver function

In some cases, both internal and external bleeding

- Laboratory findings include low white blood cell and platelet counts and elevated liver enzymes.

• **Confirmation that symptoms are caused by Ebola virus infection are made using the following diagnostic methods:**

- Antibody-capture enzyme-linked
- Immunosorbent assay (ELISA)
- Antigen-capture detection tests
- Serum neutralization test
- Reverse transcriptase polymerase chain reaction (RT-PCR) assay
- Electron microscopy
- Virus isolation by cell culture

Source :

<https://www.who.int/news-room/fact-sheets/detail/ebola-virus-disease>

Q.88) Ans: A

Exp:

- **Statement 1 is correct** : Pradhan Mantri Swasthya Suraksha Nidhi (PMSSN), a *single non-lapsable reserve fund for health* from the proceeds of Health and Education Cess levied under **Section 136-b of Finance Act, 2007**.
- PMSSN will be **utilized for the flagship schemes of the Health Ministry** — Ayushman Bharat-Pradhan Mantri Jan Arogya Yojana (AB-PMJAY), Ayushman Bharat-Health and Wellness Centres (AB-HWCs), National Health Mission, Pradhan Mantri Swasthya Suraksha Yojana (PMSSY), emergency and disaster preparedness .
- It is used for any future programme/scheme that targets to achieve **progress towards SDGs and the targets set out in the National Health Policy (NHP) 2017**.
- **Statement 2 is incorrect** : Administration and maintenance of the **PMSSN is entrusted to the Ministry of Health & Family Welfare (MoHFW)**.

Extra Edge by Only IAS

National Health Policy (NHP) 2017

- The Policy seeks to reach **everyone in a comprehensive integrated way to move towards wellness**. It aims at achieving **universal health coverage and delivering quality** health care services to all at affordable cost.
- The main objective of the *National Health Policy 2017* is to achieve the *highest possible level of good health and well-being, through preventive and promotive health care*.
- India currently has one doctor over the population of 1,445 against the WHO norm of 1:1000.
- The Combined expenditure of the Centre and States on Health is around 1.5% of India's GDP.
- It is quite below the target of 2.5% as set under the National Health Policy, 2017.
- Bhutan (2.5%) and Sri Lanka (1.6%) spend more money on health as compared to India.

Source

https://m.economictimes.com/news/economy/policy/govt-approves-creation-of-pm-swasthya-suraksha-nidhi-as-single-non-lapsable-reserve-fund-for-health/amp_articleshow/81430772.cms

Q.89) Ans: D

Exp:

- Recently , States/Union Territories (UTs) have launched Intensified Mission Indradhanush (IMI) 3.0 to cover **children and pregnant women who missed routine immunisation** during the COVID-19 pandemic.
- This is aimed to **accelerate the full immunisation of children and pregnant women through a mission mode intervention**.
- The target was to immunise all children against seven vaccine-preventable diseases namely diphtheria, pertussis, tetanus, childhood tuberculosis, polio, hepatitis B, and measles.
- **National Technical Advisory Group of India (NTAGI)**, the country's apex scientific advisory body on immunisation.
- The **Ministry of Health and Family Welfare** was responsible for the programme, with significant support from the international community.

Extra Edge by OnlyIAS

- **Universal Immunisation Programme (UIP)** is a vaccination programme launched by the Government of India in 1985.
- It became a part of Child Survival and Safe Motherhood Programme in 1992
- The **programme consists of vaccination for 12 diseases**- tuberculosis, diphtheria, pertussis (whooping cough), tetanus, poliomyelitis, measles, hepatitis B, diarrhoea, Japanese encephalitis, rubella, pneumonia (haemophilus influenzae type B) and Pneumococcal diseases (pneumococcal pneumonia and meningitis).
- **Hepatitis B and Pneumococcal** diseases were added in 2007 and 2017 respectively.
- **Four new vaccines have been introduced into the country's Universal Immunisation Programme (UIP)**, including an injectable **polio vaccine, an adult vaccine against Japanese Encephalitis and Pneumococcal Conjugate Vaccine.**

World Immunization Week

- It is a global public health campaign to raise awareness and increase rates of immunization against vaccine-preventable diseases around the world. It takes place **each year during the last week of April.**

Source :

<https://www.thehindu.com/news/national/states-roll-out-immunisation-programme-intensified-mission-indradhanush-30/article33911865.ece/amp/>

Q.90) Ans: C

Exp:

Nipah Virus

- **Nipah Virus** is an **infectious disease** that broke out in Malaysia and Singapore in 1998 and 1999.
- It first **appeared in domestic pigs** and has been found among several species of domestic animals including **dogs, cats, goats, horses and sheep.**
- The organism that **causes Nipah Virus encephalitis is an RNA or Ribonucleic acid virus of the family Paramyxoviridae, genus Henipavirus,** and is closely related to Hendra virus.
- The **virus has been listed in the World Organisation for Animal Health (OIE) Terrestrial Animal Health Code** and must be reported to the OIE (OIE Terrestrial Animal Health Code).

- The **disease spreads through fruit bats or 'flying foxes,'** of the genus Pteropus, who are natural reservoir hosts of the Nipah and Hendra viruses.
- The human infection presents as an encephalitic syndrome marked by fever, headache, drowsiness, disorientation, mental confusion, coma, and potentially death.
- **There is no specific treatment for Nipah Virus.** The primary treatment for human cases is intensive supportive care. There is possible human-to-human transmission of the disease.

Source -

<https://www.thehindu.com/opinion/editorial/nipah-amidst-a-pandemic/article36348891.ece/amp/>

Q.91) Ans: D

Exp:

- **Live-attenuated vaccine** : It uses a living but weakened version of the virus. The **measles, mumps and rubella (MMR) vaccine and the chickenpox, smallpox and shingles vaccine , and yellow fever Rotaviruses** are examples of this type of vaccine. This can be manufactured at scale. These vaccines like this may **not be suitable for people with compromised immune systems.**
- **Inactivated vaccine** : Where Disease-carrying virus or bacteria **inactivate or kill it using chemicals, heat or radiation.** This is the way the **flu and polio vaccines are made** – and vaccines can be manufactured on a reasonable scale. It requires special laboratory facilities to grow the virus or bacterium safely, and can have a relatively long production time.
- **Viral vector vaccine** : This type of vaccine uses a safe virus to deliver specific sub-parts – called **proteins – of the germ of interest so that it can trigger an immune response without causing disease.** To give instructions for making particular parts of the pathogen of interest are inserted into a safe virus. The safe virus then serves as a platform or vector to deliver the protein into the body. The **Ebola vaccine is a viral vector vaccine.**

Extra Edge by Only IAS

Messenger RNA (mRNA) :

- **These vaccines** teach our cells how to make a protein that will **trigger an immune response inside our bodies.**
- It triggers an immune response inside our bodies. That immune response, which **produces**



antibodies, is what protects us from getting infected if the real virus enters our bodies.

- The Pfizer-BioNTech and Moderna COVID-19 vaccines are **messenger RNA vaccines** – also called mRNA vaccines.
- The mRNA will enter the muscle cells and **instruct the cells' machinery to produce a harmless piece of what is called the spike protein.** The spike protein is found on the surface of the virus. After the protein piece is made, *our cells break down the mRNA and remove it.*
- Our immune system recognizes that the protein doesn't belong there. This **triggers our immune system to produce antibodies and activate other immune cells to fight off what it thinks is an infection.**
- Ultimately, our bodies have learned how to protect against future infection from the virus

Source - <https://www.who.int/news-room/feature-stories/detail/the-race-for-a-covid-19-vaccine-explained>

lactating women, and adolescent girls. This mission was aided by International Bank for Reconstruction and Development and other multilateral development agencies. **The 50 % fund will be provided by the World Bank to implement the mission.**

The mission sets a target to reduce stunting, under-nutrition, and low birth weight by 2 per cent per annum, and anaemia by 3 percent annually.

Source:

<https://indianexpress.com/article/india/govt-launches-nutrition-mission-aimed-at-10-crore-beneficiaries-4964081/lite>

Q.93) Ans: D

Exp:

National Digital Health Mission

- The **National Digital Health Mission** had been **announced by PM Modi** on August 15, 2020.
- The mission aims to **create an integrated healthcare system linking practitioners with the patients digitally** by giving them access to real-time health records.
- Currently, the Pradhan Mantri Digital Health Mission (PM-DHM) is being implemented in a pilot phase in 6 UTs.
- Through this mission, a **unique digital health ID will be provided to the people**, which will contain all the health records of the person.
- The nationwide rollout of PM-DHM coincides with **National Health Authority (NHA) celebrating the third anniversary of Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB PM-JAY).**
- The **key components of PM-DHM include a health ID for every citizen** that will also work as their health account, to which personal health records can be linked and viewed with the help of a mobile application.
- The mission **can access and exchange longitudinal health records of citizens with their consent.**
- The Healthcare Professionals Registry (HPR) and Healthcare Facilities Registries (HFR) will act as a repository of all healthcare providers across both modern and traditional systems of medicine.

Source: <https://www.livemint.com/news/india/pm-modi-to-launch-national-digital-health-mission-nationwide-on-monday-key-points/amp-11632655217995.html>

Q.94) Ans: A

Q.92) Ans: C

Exp:

- **Statement 1 is correct** : The **National Nutrition Mission**, launched in **2018**. It is the Government of India's flagship programme to **improve nutritional outcomes for children, pregnant women and lactating mothers.** National Nutrition Strategy prepared by the **NITI Aayog** with the goal of attaining **"Kuposhan Mukta Bharat" or malnutrition-free India, by 2022.** Mission will be executed with the **Ministry of Women and Child Development (WCD)** as the nodal ministry along with the Ministry of Drinking Water and Sanitation and the Ministry of Health and Family Welfare.
- **Statement 2 is incorrect** : The Centre's budgetary support is further divided into **60:40 between the Centre and the States, 90:10 for the north-eastern region and the Himalayan States and 100% for the Union Territories (UTs)** without a legislature. The mission aims to **bring down the stunting percentage as per the National Family Health Survey-4 (2014-15) from 38.4 per cent to 25 per cent by 2022.**
- **Statement 3 is correct** : The **National Nutrition Mission** aimed at **10 crore beneficiaries, mainly children up to the age of 6 years, pregnant and**



Exp:

- Recently, the **World Bank released the biennial 'Poverty and Shared Prosperity Report 2020'**.
- Highlights of report -**
 - Global poverty had **declined at the rate of around 1 percentage point per year** between 1990 and 2015.
 - In two-and-a-half decades (1990-2015), the **extreme poverty rate declined by 26 percentage points** and it dropped to 10 per cent from nearly 36 per cent.
 - But, In 2020 , Global extreme poverty is expected to rise for the first time in 20 years because of the disruption caused by COVID-19.**
 - This will exacerbate the impact of conflict and climate change, which are already slowing down poverty reduction.
 - The pandemic may push another 88 million to 115 million into extreme poverty or having to live on less than \$1.50 per day, resulting in a total of 150 million such individuals.

Extra Edge by Only IAS

Poverty and Shared Prosperity Report and India

- India, along with Nigeria**, is considered to have the largest number of the poor in the world.
- India tops the global list in terms of absolute number of poor, according to the last national survey of 2012-13.
- The **country accounted for 139 million** of the total 689 million people living in poverty in 2017.
- India joins the ranks of countries termed as 'conflict-affected' and 'fragile' in World Bank terminologies.*

Source :

<https://www.thehindu.com/news/international/coronavirus-pandemic-could-push-150-million-into-extreme-poverty-says-world-bank/article32793466.ece/amp/>

Q.95) Ans: D

Exp:

- Havana Syndrome** refers to a **set of mental health symptoms that may involve hearing certain sounds without any outside noise being present,**

nausea, vertigo and headaches, memory loss and issues with balance.

- As per study by scientists in the US and medical examination of the victims , the victims may have been subjected to **high-powered microwaves that either damaged or interfered with the nervous system.**
- It can occur due to pressure inside the brain that generates the feeling of a sound being heard. **Greater exposure to high-powered microwaves** is said not only to interfere with the **body's sense of balance** but also impact memory and cause permanent brain damage.

Extra Edge by Only IAS

Stockholm syndrome

- It is a condition in which **hostages develop a psychological bond with their captors during captivity.**
- Stockholm Syndrome occurred due to a specific set of circumstances, namely the **power imbalances contained in hostage-taking, kidnapping, and abusive relationships.**

Source :

<https://indianexpress.com/article/explained/cia-officer-on-india-trip-reports-havana-syndrome-symptoms-causes-7525418/lite/>

Q.96) Ans: C

Exp:

- Both statements are correct** : The Food Safety and Standards Authority of India (FSSAI) has capped the amount of trans fatty acids (TFA) in oils and fats to 3% for 2021 and 2% by 2022 from the current permissible limit of 5%.
- The revised regulation applies to edible refined oils, vanaspati (partially hydrogenated oils), margarine, bakery shortenings, and other mediums of cooking such as vegetable fat spreads and mixed fat spreads.
- The **WHO has also called for global elimination of trans fats by 2023.** As per the World Health Organisation, approximately 5.4 lakh deaths take place each year globally because of intake of trans fatty acids.
- Industry players have to take a pledge , they would comply with **WHO's call for action to reduce TFA by 3% by 2021** allowing them three years to comply with the latest regulation.

- As per WHO, increased intake of trans-fat – more than 1% of total energy intake- is associated with increased risk of coronary heart disease mortality.
- Highest levels of TFA are present in vanaspati, commonly used as vegetable ghee for household cooking, and consumed in bakery items, sweets, street food, and packaged foods.

Extra Edge by Only IAS

Food Safety and Standards Authority of India (FSSAI)

- It is a statutory body established under the Ministry of Health & Family Welfare, Government of India.
- The FSSAI has been established under the Food Safety and Standards Act, 2006, which is a consolidating statute related to food safety and regulation in India.
- FSSAI is responsible for protecting and promoting public health through the regulation and supervision of food safety.
- The FSSAI is headed by a non-executive Chairperson, appointed by the Central Government, either holding or has held the position of not below the rank of Secretary to the Government of India.
- The FSSAI has its headquarters at New Delhi. The authority also has 6 regional offices located in Delhi, Guwahati, Mumbai, Kolkata, Cochin, and Chennai.

Source :

<https://www.thehindu.com/sci-tech/health/fssai-slashes-limit-for-trans-fat-levels-in-foods/article33486031.ece/amp/>

Q.97) Ans: D

Exp:

- Both statements are incorrect. Cholesterol travels through the blood on proteins called "lipoproteins."
- Two types of lipoproteins carry cholesterol throughout the body.

LDL (low-density lipoprotein)	HDL (high-density lipoprotein)
LDL (low-density lipoprotein), also called "bad" cholesterol, makes up most of	HDL (high-density lipoprotein), or "good" cholesterol, absorbs cholesterol and carries it back

your body's cholesterol. High levels of LDL cholesterol raise your risk for heart disease and stroke.

to the liver. High levels of HDL cholesterol can lower your risk for heart disease and stroke.

- Recently, The Food Safety and Standards Authority of India (FSSAI) has capped the amount of trans fatty acids (TFA) in oils and fats to 3% for 2021 and 2% by 2022 from the current permissible limit of 5%.
- The revised regulation applies to edible refined oils, vanaspati (partially hydrogenated oils), margarine, bakery shortenings, and other mediums of cooking such as vegetable fat spreads and mixed fat spreads.
- The WHO has also called for global elimination of trans fats by 2023. As per the World Health Organisation, approximately 5.4 lakh deaths take place each year globally because of intake of trans fatty acids.

Source :

https://www.cdc.gov/cholesterol/ldl_hdl.htm

Q.98) Ans: A

Exp:

'Draft Food Safety and Standards (Amendment) Bill, 2020'

- The Ministry of Health & Family Welfare released the draft bill that seeks to amend the Food Safety and Standards Act, 2006 to revamp the Food Safety and Standards Authority of India (FSSAI).
- The aim of the bill is to revamp the functioning and jurisdiction of FSSAI so this bill gives more powers, enhancing penalties imposed for violations and simplifying processes.
- Jurisdiction of FSSAI extended to include animal feed. More powers to the Food Safety and Standards Authority of India. The jurisdiction of FSSAI extended to the animal feed industry.
- The Food Authority can prescribe a mechanism for handling food safety incidents.
- The FSSAI will specify standards for food contact material i.e., food packaging material. The food packaging material is mentioned as "food contact material" in the bill.
- New definitions for Proprietary and Novel food have been included in the Act.

- The Food Authority will set up the Food Safety and Standards Authority of India Fund.

Extra Edge by Only IAS

The Food Safety and Standards Authority of India

- The Food Safety and Standards Authority of India was established under the Food Safety and Standards Act, 2006.
- It is a statutory body established under the Ministry of Health & Family Welfare, Government of India.
- The FSSAI has been established under the Food Safety and Standards Act, 2006, which is a consolidating statute related to food safety and regulation in India.
- FSSAI is responsible for protecting and promoting public health through the regulation and supervision of food safety.
- The FSSAI has its headquarters at New Delhi. The authority also has 6 regional offices located in Delhi, Guwahati, Mumbai, Kolkata, Cochin, and Chennai.

Source

<https://www.foodsafetymantra.com/regulatory-update/consumer-products/the-food-safety-and-standards-amendment-bill-2020/>

Q.99) Ans: D

Exp:

- The National Tuberculosis Elimination Program (NTEP) is the Public Health initiative of the Government of India.
- It functions as a flagship component of the National Health Mission (NHM) and provides technical and managerial leadership to anti-tuberculosis activities in the country.
- National Strategic Plan 2017-25, the program has a vision of achieving a "TB free India", with a strategy under the broad themes of "Prevent, Detect, Treat and Build pillars for universal coverage and social protection".
- The program provides various free of cost, quality tuberculosis diagnosis and treatment services across the country through the government health system.
- Statement 1 is correct : At national level the program is led by the Central TB Division under the Ministry of Health and Family Welfare.
- Statement 2 is correct

WHO End TB Strategy: 80% reduction in the TB incidence rate (new and relapse cases per 100 000 population per year) by 2030.

India's goal of eliminating the TB disease by 2025, five years ahead of the Sustainable Development Goals target.

- Statement 3 is correct : Moscow Declaration to End TB: It is a commitment to increase multi-sectoral action and enhance accountability in the global TB response towards ending tuberculosis (TB) by 2030.

Source - <https://www.thehindu.com/sci-tech/science/tuberculosis-control-programme-renamed-to-focus-on-elimination/article30586196.ece/amp/>

Q.100) Ans: A

Exp:

- Statement 1 is correct : 'The Cancer Genome Atlas (TCGA) is a cancer genomics program of the US-India which began in 2006 bringing together researchers from diverse disciplines and multiple institutions.
- TCGA is a landmark project which has provided publicly available data sets to help improve diagnostic methods and treatment standards to prevent cancer.
- Statement 2 is incorrect : TCGA, a collaboration between the National Cancer Institute and the National Human Genome Research Institute, has generated comprehensive multi-dimensional maps of the key genomic changes in 33 types of cancer.
- Similarly, The establishment of an 'Indian Cancer Genomics Atlas (ICGA)' has been initiated by a consortium of key stakeholders in India led by CSIR.
- Through this project several government agencies, cancer hospitals, academic institutions and the private sector work together to combat cancer.



Extra Edge by Only IAS

Cancer

- Cancer is a group of diseases involving abnormal cell growth with the potential to invade or spread to other parts of the body
- A neoplasm and malignant tumor are other common names for cancer.
- Lung, prostate, colorectal, stomach and liver cancer are the most common types of cancer in men, while breast, colorectal, lung, cervical and thyroid cancer are the most common among women.